Moon Formation via Triple Phase Transition in the Differentiating Proto-Earth

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Abstract. The origin of the Moon remains one of the unresolved problems of planetary science. The canonical giant-impact model faces growing geochemical difficulties: it does not naturally predict the near-perfect Earth–Moon isotopic identity, the crustal dichotomy, or the ≈ 350 Myr delay of the terrestrial dynamo. This work proposes an alternative grounded in a necessary thermodynamic observation: every Earth-mass planet emerges from accretion in a state of near-total melting of the silicate mantle, the accretion energy exceeding the total melting energy by a factor ≈ 155 (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). The proto-Earth is therefore a rapidly rotating magmatic body ($T_{\text{rot}} \approx 3.5$ h) with no stabilizing satellite.
In the Hadean context — a T-Tauri Sun whose coronal mass ejections are 10 to 100 times more intense than today's (Feigelson & Montmerle, 1999; Airapetian et al., 2016), a dense silicate atmosphere at 2,000–4,000 K acting as a thermal blanket (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013), continuous planetesimal bombardment (Agnor et al., 1999; Kokubo & Genda, 2010), and active short-lived radioactivity (Urey, 1955; Huss et al., 2009) — the proto-Earth's axis undergoes large-amplitude free nutation within $[40°, 70°]$ in its own co-rotating frame (Laskar et al., 1993a; Laskar & Robutel, 1993b; Touma & Wisdom, 1993; Li & Batygin, 2014). This is *not* astronomical obliquity relative to a nonexistent Hadean ecliptic; it is a purely internal geometric oscillation of the fluid body.
In this context, a single engine — the progressive segregation of Fe-Ni toward the forming core — governs three coupled transitions. The first is a rheological transition: the silicate magma acquires a Bingham-Herschel yield stress (Bingham, 1922; Herschel & Bulkley, 1926) and structures a Coherent Magmatic Torus (CMT) within the intertropical band $|\phi| < 30°$. The second is a mechanical transition: the CMT undergoes two to three episodes of cohesive hypersonic ejection beyond the Roche radius, governed by a double potential well and the stochastic Kramers crossing rate (Kramers, 1940). The third is a magnetic transition: the terrestrial dynamo switches on ≈ 350 Myr after the ejections cease, consistent with the paleomagnetic data from Jack Hills zircons (Tarduno et al., 2025).
The central equation yields an ejected mass of 2.2–4.0 $\times 10^{22}$ kg per episode; with $N = 2$–3 episodes and a capture efficiency of 0.70, the lunar mass is reconstituted. The mechanism requires no external impactor and satisfies the constraints of isotopic identity, iron depletion, and angular momentum as direct mechanical consequences. The central prediction is one or more seismic interfaces between 200 and 530 km depth, with an impedance contrast $|R| \in [0.01, 0.04]$. This prediction is testable by **Chang'e 7** (South Pole, August 2026), **FSS** (Farside Seismic Suite), **LEMS** (Lunar Environment Monitoring Station), and **Artemis III** (2028–2029). Preliminary observational support is provided by Chang'e-6 samples (Yue et al., 2026; Xu et al., 2025): norites dated at 4247 ± 5 Ma (consistent with Episode 1) and anomalously high Fe/Mn ratios in deep olivines, both consistent with predictions P3 and P6 of this work. All hypotheses are explicit and hierarchized. Limitations are acknowledged. The validation program is defined and sequenced.

Interactive simulation: https://orion4622.github.io/moon-formation-triple-phase-transition/Animation_Formation_de_la_Lune_v2.html

Keywords: origin of the Moon · Triple Phase Transition · chaotic axial oscillation · Hadean proto-Earth · Bingham-Herschel · Coherent Magmatic Torus · double potential well · Kramers rate · Mathieu resonance · parametric elliptical instability · seismic interface · Hadean dynamo · falsifiable predictions · South Pole–Aitken · Chang'e-6.

Epistemic Posture

This work proposes a theory and submits it to the scrutiny of the scientific community. The phrasing deliberately reflects this posture: the theory describes, predicts, and constrains — it does not assert absolute certainties. All hypotheses are numbered and hierarchized. All limitations are acknowledged.

On numerical simulations: A complete 3D SPH validation with Bingham-Herschel rheology, radiative cooling, Fe-Ni segregation, and $\varepsilon \neq 0$ geometry exceeds the capabilities of an isolated independent researcher and belongs to an institutional team with HPC resources. The TPT does not require such a simulation to be evaluated; it will be judged primarily on its observational predictions (Sections 13 and 14). The final arbitration rests with forthcoming lunar seismic data (Chang'e 7, August 2026) and, in a second stage, with three-dimensional numerical simulations once they become available.

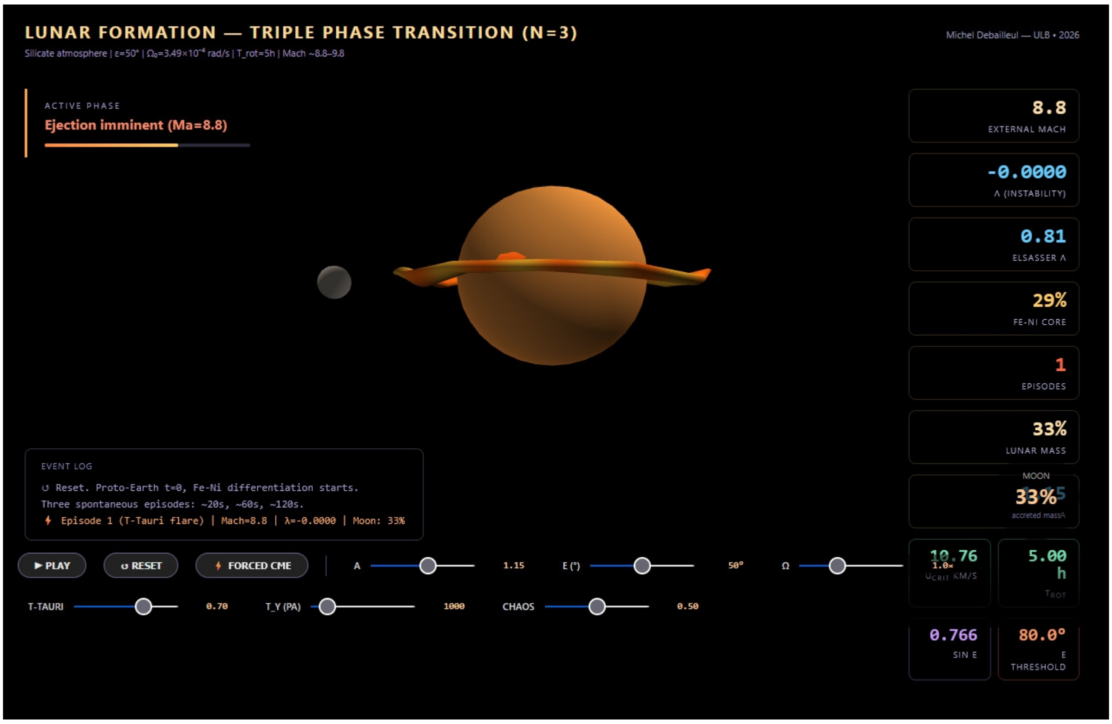

Figure 1: Screenshot of the interactive simulation accompanying this work (link above), illustrating the Coherent Magmatic Torus (CMT) during ejection around the proto-Earth, with the real-time dynamic dashboard of parameters (external Mach number, λ instability, Elsasser number, Fe-Ni core fraction, accreted lunar mass).

1 The Hadean Inferno: Physical Context

Before entering into the physics of the Triple Phase Transition, it is essential to understand the framework in which it operates. This framework is not a quiet void: it is one of the most energetically violent periods in the history of the Solar System.

1.1 The Dynamical Context of Terminal Accretion

The Hadean period (4.568–4.4 Ga) is characterized by a set of intense physical processes that define the initial conditions and continuous forcings of the system:

- **Continuous planetesimal accretion**: the protoplanetary disk remains populated with planetesimals and embryos, providing constant bombardment (Agnor et al., 1999; Kokubo & Genda, 2010).
- **Rapid rotation**: the proto-Earth rotates with ≈ 3.5 h, a regime in which Coriolis and centrifugal effects dominate internal dynamics.
- **Absence of a tidal stabilizer**: without a Moon, obliquity is free to evolve over a wide chaotic range.
- **A young, active Sun**: the T-Tauri Sun irradiates intensely and produces coronal mass ejections (CMEs) that perturb the system (Feigelson & Montmerle, 1999; Airapetian et al., 2016).
- **Short-lived radioactivity**: the isotopes ^{26}Al and ^{60}Fe are still active, injecting heat into the mantle (Urey, 1955; Huss et al., 2009).
- **Dense silicate atmosphere**: an envelope of rock vapor at $T \sim 2,000$–$4,000$ K and $P \sim 10^5$–10^7 Pa constitutes a thermal blanket and a lateral confinement agent (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013).

The combination of these processes — not any single one of them — defines the physical environment in which the TPT operates. The T-Tauri Sun, although important, is only one ingredient among others in this dynamical context.

1.2 The Silicate Atmosphere: Thermal Blanket and Lateral Confinement

At $t = 0$, the proto-Earth is enveloped in a dense silicate atmosphere — rock vapor, gaseous SiO, Mg, and Fe — at temperatures of 2,000 to 4,000 K and pressures of 10^5–10^7 Pa. This cocoon plays two critical roles in the TPT.

First role: thermal insulation. The silicate atmosphere has high infrared opacity. It reduces surface radiative cooling by a factor of 10^4 to 10^5 (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013). Without this atmosphere, the proto-Earth would have cooled within $\sim 10^4$ years. With it, the melting window extends to 30–50 Myr — exactly the timescale of core formation (Kleine et al., 2002; Yin et al., 2002).

This is not an ad hoc hypothesis: it is the physical consequence of a rock-vapor atmosphere at $2,000$–$4,000$ K. The same mechanism operates in magma ocean models (Elkins-Tanton, 2008; Lebrun et al., 2013).

Second role: lateral confinement. The confining pressure ($P \approx 10^5$–10^7 Pa) opposes lateral expansion of the CMT flow during ejection. It guarantees the cohesion of the ejected sheet. The high sound speed in this torrid atmosphere ($c_{\text{sound}} \approx 1,500$ m s^{-1} at $T_{\text{atm}} \approx 3,000$ K) reduces the effective Mach number of the flow, improving cohesion relative to ejection into vacuum.

Established Result

The silicate atmosphere is a thermal blanket. Without it, the melting window would close within $\sim 10^4$ years. With it, the window extends to 30–50 Myr (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013), matching the timescale of core formation (Kleine et al., 2002; Yin et al., 2002).

1.3 Continuous Planetesimal Bombardment

The protoplanetary disk remains populated with planetesimals and embryos during the active TPT window (Agnor et al., 1999; Kokubo & Genda, 2010; Chambers, 2001). Each impact: (1) re-injects thermal energy into the mantle, sustaining melting; (2) generates a spherical bulk pressure wave propagating through the entire molten mass, perturbing $U(r)$ and contributing stochastically to Kramers barrier crossings; (3) delivers an impulsive torque on the spin vector, resetting the axial oscillation.

1.4 Short-Lived Radioactivity: Internal Heat Source

The isotopes ^{26}Al (half-life 0.72 Myr) and ^{60}Fe (half-life 2.6 Myr) are still active during the first few million years (Urey, 1955; Huss et al., 2009), injecting $\approx 3 \times 10^{29}$ J of additional heat into the mantle and contributing to sustained melting throughout the TPT window.

1.5 Hadean Obliquity $\varepsilon \in [40°, 70°]$: A Fundamental Geometric Clarification

The range $\varepsilon \in [40°, 70°]$ is not a free parameter. It is constrained by five independent, convergent physical arguments. Before listing them, a fundamental geometric clarification is needed.

Geometric Clarification — Read First

In this work, ε denotes the instantaneous angle between the rotation vector $\mathbf{\Omega}$ and the principal axis of inertia of the Maclaurin spheroid, **defined in the body's co-rotating frame**. This is *not* astronomical obliquity relative to the ecliptic plane: this concept has no physical meaning for the Hadean proto-Earth, which has neither a fixed reference ecliptic nor a Moon. It is a purely internal geometric angle describing the *oscillation* — the free nutation — of the rotation axis relative to the body's own shape. The Hadean intertropical band $|\phi| < 30^\circ$ is defined perpendicular to the *instantaneous* Hadean rotation axis, not relative to a fixed geographic frame.

Argument 1 — Absence of a tidal stabilizer (fundamental). The proto-Earth has no Moon. Without a massive satellite, there is no lunar tidal torque to damp the precession of the spin axis and keep it near a stable equilibrium. This argument is logically prior to all others: it is precisely the Moon whose formation we seek to explain, so invoking it as a stabilizer would be circular. The absence of a tidal stabilizer is therefore a necessary, not contingent, condition of the Hadean proto-Earth.

Argument 2 — Laskar et al. (1993): a chaotic zone without a satellite. Laskar et al. (1993a) showed that without its Moon, Earth's obliquity would evolve chaotically over a zone extending from nearly 0° to $\approx 85^\circ$. Although this result applies strictly to a present-day Earth rotating at ≈ 24 h, it establishes the general principle: *the Moon is the stabilizer, and without it, large obliquity excursions are the norm*. For the Hadean proto-Earth, this principle applies *a fortiori*.

Argument 3 — Rapid rotation at $T_{\text{rot}} = 3.5$ h. The precession constant satisfies $\alpha_{\text{prec}} \propto \omega$. At $T_{\text{rot}} = 3.5$ h, the precession frequency is $\approx 7\times$ higher than at 24 h, pushing the secular spin-orbit resonances identified by Laskar et al. (1993a) out of reach. Li & Batygin (2014), working with a rapidly rotating moonless Earth, found that obliquity remains chaotic but *diffuses slowly* over a wide range. Rapid rotation does not stabilize the axis — it simply changes the character of the chaos from resonant to diffusive.

Argument 4 — Continuous planetesimal bombardment. Each major impact during terminal accretion (Agnor et al., 1999) abruptly resets the spin vector. The relaxation time toward axial alignment for an isolated fluid body,

$$\tau_{\text{relax}} \sim \frac{\eta\, R^3_{\text{eq,proto}}}{G\, M^2_\oplus\, f} \approx 10^6\,\text{yr} \quad (\eta \sim 10^{1\text{–}3}\,\text{Pa s},\ f \approx 0.107), \tag{1}$$

far exceeds the interval between impacts (10^3–10^5 yr). Misalignment is continuously maintained: $\tau_{\text{pert}} \ll \tau_{\text{relax}}$.

Argument 5 — Torques from T-Tauri CMEs. As described in Section 5.3, the impulsive torques from CMEs act on timescales of days to weeks, far shorter than τ_{relax}. They constitute a continuous stochastic source of obliquity perturbation (Feigelson & Montmerle, 1999; Airapetian et al., 2016).

Physical consequence. The range $\varepsilon \in [40°, 70°]$ is not an ad hoc hypothesis. It is the predictable consequence of the dynamics of a rapidly rotating fluid body, deprived of a tidal stabilizer, subjected to continuous stochastic perturbations. The simulations of Laskar et al. (1993a) for a moonless Earth, and of Li & Batygin (2014) for a rapidly rotating moonless Earth, converge on this range — the same one adopted by Laskar et al. (1993a) and Touma & Wisdom (1993).

Geometric consequence for the Coriolis force. In this oblique geometry, the Coriolis force decomposes into radial and horizontal components (Section 5.2). The radial component $f_{\text{rad}} = 2\Omega \sin\varepsilon \cdot U$ is maximal for $\varepsilon \in [40°, 70°]$.

Established Result

Five convergent arguments support $\varepsilon \in [40°, 70°]$. The absence of a tidal stabilizer is logically necessary. Arguments 2 through 5 provide independent physical justification for a large, sustained misalignment. The range $[40°, 70°]$ is adopted as the physically motivated Hadean range; the mechanism operates for any $\varepsilon > 45°$ (Section 5.2).

1.6 Relaxation Time and Rheological Reinforcement

The relaxation time toward axial alignment for an isolated fluid body is given by Eq. (1). Bingham-Herschel rheology further slows this relaxation. The rheological response time to a stress perturbation is:

$$\tau_{\text{BH}} \sim \frac{\tau_y}{\mu \dot{\gamma}_{\text{typ}}} \approx 10^2\text{–}10^4\ \text{s}, \tag{2}$$

where μ is the dynamic viscosity and $\dot{\gamma}_{\text{typ}}$ the typical shear rate. This time is much shorter than τ_{relax}, meaning that mass redistribution is locally frozen by the yield stress, further delaying axial alignment.

2 Initial State of the Proto-Earth

2.1 Near-Total Melting: A Necessary Thermodynamic Consequence

The formation of an Earth-mass planet releases an enormous amount of gravitational energy. In the homogeneous-sphere approximation (Safronov, 1969), the total accretion energy is:

$$E_{\text{grav}} = \frac{3\,G\,M_\oplus^2}{5\,R_{\text{eq,proto}}} \approx 2.49 \times 10^{32}\,\text{J}, \tag{3}$$

where G is the gravitational constant, $M_\oplus$ the Earth's mass, and $R_{\text{eq,proto}} \approx 7.4$ Mm the equatorial radius of the undifferentiated proto-Earth.

Established Result

Equation (3) **is a thermodynamic bound, not a model of instantaneous formation.** It calculates the *total* gravitational energy released during accretion, regardless of its duration. This is the standard approach in the literature (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). The manuscript explicitly cites Kleine et al. (2002); Yin et al. (2002) as evidence that accretion took ~ 30 Myr.

The energy required to melt the entire silicate mantle (Stixrude & Lithgow-Bertelloni, 2014):

$$E_{\text{fus}} \approx M_{\text{mantle}} \times L_{\text{fus}} \approx 4 \times 10^{24}\,\text{kg} \times 4 \times 10^{5}\,\text{J}\,\text{kg}^{-1} \approx 1.6 \times 10^{30}\,\text{J}, \tag{4}$$

where M_{mantle} is the mass of the silicate mantle and L_{fus} the specific latent heat of fusion.

Established Result

Energy ratio — a consensual result.

$$E_{\text{grav}}/E_{\text{fus}} \approx 155. \tag{5}$$

Established by Solomatov (2000), Elkins-Tanton (2012), and Rubie et al. (2015). The accretion energy exceeds the total melting energy by two orders of magnitude.

Why Melting Was Sustained Despite Radiative Cooling

Accretion took ~ 30 Myr (Kleine et al., 2002; Yin et al., 2002). During this period, radiative cooling occurred. Four independent mechanisms explain why melting was sustained despite this cooling.

Mechanism 1 — The silicate atmosphere as a thermal blanket. The dense silicate atmosphere ($P \approx 10^5$–10^7 Pa, $T \approx 2{,}000$–$4{,}000$ K) has high infrared opacity. It reduces surface radiative cooling by a factor of 10^4 to 10^5 (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013). The effective cooling time goes from $\sim 10^4$

years to 30–50 Myr.

Mechanism 2 — Additional heat sources. Four heat sources prolong melting beyond the initial accretion energy:

- **Fe-Ni segregation heat:** $\approx 3 \times 10^{30}$ J (Flasar & Birch, 1973; Rubie et al., 2003).
- **Short-lived radioactivity:** ^{26}Al (half-life 0.72 Myr) and ^{60}Fe (half-life 2.6 Myr) inject $\approx 3 \times 10^{29}$ J (Urey, 1955; Huss et al., 2009).
- **Continuous planetesimal bombardment:** re-injection of thermal energy (Agnor et al., 1999; Kokubo & Genda, 2010).
- **T-Tauri irradiation:** maintains surface temperatures above 2,000 K (Feigelson & Montmerle, 1999; Airapetian et al., 2016).

Mechanism 3 — The TPT does not require melting of the entire mantle. The Triple Phase Transition requires melting only within the Hadean intertropical band $|\phi| < 30^\circ$ (upper and mid mantle). The factor of 155 thermodynamically guarantees this partial melting, even if the deep mantle was not entirely molten (Nomura et al., 2011).

Mechanism 4 — The active TPT window is 30–50 Myr. The TPT occurs *during* the 30–50 Myr window in which melting is sustained by the mechanisms above. Core formation over ~ 30 Myr (Kleine et al., 2002; Yin et al., 2002) is subsequent to or simultaneous with this, not a contradiction.

Established Result

Near-total melting is the mandatory starting point. The objection based on the ~ 30 Myr accretion timescale does not contradict the mechanism: the TPT occurs *during* the melting window sustained by the silicate atmosphere, the additional heat sources, and continuous bombardment.

Hypothesis H0 — Total volumetric melting. At the onset of Fe-Ni segregation, the entire silicate mantle lies above the liquidus, with no stable internal solid interface during the active window 4.568–4.4 Ga. This is a self-gravitating fluid volume, not a magma ocean resting on a substrate: the dynamics, rheology, and response to perturbations are physically distinct.

Caveat: Nomura et al. (2011) suggest that the deep lower mantle may not have been entirely molten owing to the rise of the solidus with pressure. The TPT requires melting only within the Hadean intertropical band $|\phi| < 30^\circ$ (upper and mid mantle), which the factor of 155 thermodynamically guarantees. The adiabatic temperature profile (Appendix A) confirms melting throughout the mantle.

2.2 Initial Rotation Period: $T_{\mathrm{rot}} \approx 3.5$ h

The proto-Earth's initial rotation period is not a free parameter. Three convergent physical mechanisms constrain it to $T_{\mathrm{rot}} \approx 2.5$–$3.5$ h.

1. T-Tauri magnetic torque. The young, rapidly rotating Sun (≈ 3 days, Bouvier et al. 2014) magnetically couples to the inner protoplanetary disk (Königl, 1991; Shu et al., 1994), transferring angular momentum to material in the inner orbit and accelerating the proto-Earth.

2. Angular momentum conservation during contraction. Contraction from an initial radius $R_0 \approx 2\,R_\oplus$ to the final radius conserves angular momentum:

$$\frac{T_{\mathrm{rot}}^{(f)}}{T_{\mathrm{rot}}^{(0)}} = \left(\frac{R_\oplus}{R_0}\right)^2 \approx \frac{1}{4}. \tag{6}$$

Starting from a median initial period $T_{\mathrm{rot}}^{(0)} \approx 14$ h, one obtains $T_{\mathrm{rot}}^{(f)} \approx 3.5$ h.

3. Skater effect: Fe-Ni segregation. The migration of Fe-Ni toward the forming core reduces the moment of inertia. With $I_f / I_0 \approx 0.82$ (Rubie et al., 2015):

$$T_{\mathrm{rot}}^{(f)} \approx 0.82 \times T_{\mathrm{rot}}^{(0)}, \tag{7}$$

contributing an additional $\approx 18\%$ acceleration.

Established Result

$T_{\mathrm{rot}} \approx 3.5$ **h — consistent with community consensus.** N-body simulations give rotation periods between 2 and 8 h, with a median of 3–5 h (Agnor et al., 1999; Kokubo & Genda, 2010; Chambers, 2013). The value 3.5 h is the same as that adopted by Ćuk & Stewart (2012) and Canup (2012) in the giant-impact model.

The Coriolis compensation parameter b' is a simple ratio between the critical ejection velocity U_{crit} and the geostrophic equilibrium velocity of the CMT. Fe-Ni segregation increases b'; ejections decrease it. The equilibrium point $b' = 1$ is a stable attractor, which naturally selects $T_{\mathrm{rot}} \approx 3.5$ h and $\varepsilon \approx 55°$. The complete derivation is in Appendix B.

2.3 Geometry of the Proto-Earth: The Maclaurin Spheroid

A rapidly rotating fluid mass takes the shape of a Maclaurin spheroid (Chandrasekhar, 1969). The equatorial radius is $R_{\mathrm{eq,proto}} \approx 7.4$ Mm, combining reduced density (+4%), thermal expansion (+7.5%), and equatorial bulging (+7.5%). The dynamical flattening:

$$J_2 \approx \frac{5}{4} \cdot \frac{\Omega^2 R_\oplus}{g_0} \approx 0.101, \tag{8}$$

where g_0 is the mean surface gravity, gives an equatorial bulge $\delta R \approx 642$ km. Effective gravity varies with latitude ϕ as:

$$g_{\text{eff}}(\phi) = g_{\text{eff}}^{(0)} \left(1 + \tfrac{3}{2} J_2 \sin^2 \phi\right) - \Omega^2 \, r(\phi) \cos^2 \phi, \tag{9}$$

where $r(\phi)$ is the local radius. Effective gravity is minimal within the band $|\phi| < 30°$: this is the geometric attractor of the CMT.

The free (Euler) nutation period is:

$$\omega_E = \Omega f \approx 3.74 \times 10^{-5} \,\text{rad}\,\text{s}^{-1}, \qquad T_{\text{nut}} \approx 46.7\,\text{h}. \tag{10}$$

This frequency plays a central role in the parametric elliptical instability (Section 5.5).

3 The Intertropical Band $|\phi| < 30°$: The Ejection Site

A fundamental and often misunderstood point: the CMT is not an equatorial structure. It is a coherent zonal band extending over $\pm 30°$ of oblique latitude — roughly one third of the proto-Earth's surface. This extent results from three independent convergent mechanisms.

Mechanism 1 — Maclaurin geometry. Effective gravity is minimal within the band $|\phi| < 30°$ (Eq. 9). Molten material accumulates there by centrifugal effect, independently of any other argument. This is a consequence of the ellipsoidal shape imposed by rapid rotation.

Mechanism 2 — The geostrophic inverse cascade. The CMT's Rossby number is deeply geostrophic:

$$\text{Ro} = \frac{U_{\text{turb}}}{2\,\Omega\, R_{\text{eq,proto}}} \approx 1.35 \times 10^{-4} \ll 1, \tag{11}$$

In this regime, turbulent energy cascades toward large scales (inverse cascade) (Rhines, 1975). The Rhines scale:

$$L_\beta = \sqrt{\frac{U_{\text{turb}}}{\beta}}, \qquad \beta = \frac{2\Omega \cos \varepsilon}{R_{\text{eq,proto}}}, \tag{12}$$

gives $L_\beta \approx 0.52\, R_\oplus$ — corresponding to the harmonic mode $\ell = 1, m = 1$: the largest coherent structure possible on a sphere. The CMT is the statistical attractor of this cascade (Rhines, 1975; Sukoriansky et al., 2007).

Mechanism 3 — The radial Coriolis force. $f_{\rm rad} = 2\Omega U \sin\varepsilon$ concentrates the flow within the oblique intertropical band and keeps it coherent there.

These three mechanisms are independent and converge on the same band $|\phi| < 30°$. Their simultaneous convergence is an argument for the theory's structural robustness.

4 The CMT as a Purely Toroidal Flow: Answering the Taylor-Proudman Objection

Any geophysical fluid dynamicist confronted with the CMT will naturally raise the following objection: in a rapidly rotating fluid (Ro $\ll 1$), the Taylor-Proudman theorem imposes $(\boldsymbol{\Omega} \cdot \nabla)\boldsymbol{u} = 0$, generating column-like structures parallel to $\boldsymbol{\Omega}$, rather than a single oblique band $|\phi| < 30°$. The objection is legitimate and deserves a precise answer.

4.1 The Poloidal/Toroidal Decomposition

Any velocity field within a spherical shell decomposes into a poloidal part (radial and meridional components) and a toroidal part (purely azimuthal component):

$$\boldsymbol{u} = \boldsymbol{u}_{\rm pol} + \boldsymbol{u}_{\rm tor}, \qquad \boldsymbol{u}_{\rm tor} = \nabla \times (\psi \hat{\boldsymbol{r}}).$$

The Taylor-Proudman theorem, in its classical form, constrains the poloidal field (which carries vorticity along $\boldsymbol{\Omega}$) and produces Taylor columns. It does not constrain the purely toroidal field.

For a purely azimuthal flow independent of longitude ($u_r = u_\theta = 0$ on average, $u_\phi = u_\phi(r, \phi)$), the Taylor-Proudman constraint is automatically satisfied:

$$(\boldsymbol{\Omega} \cdot \nabla) u_{\rm tor} = 0 \tag{13}$$

for any toroidal field, with no additional condition. The balance is cyclostrophic, not geostrophic: the centrifugal pressure gradient balances the Coriolis force within the azimuthal plane.

4.2 Why Curved Columns Do Not Extend Beyond the Active Band

In a spherical shell rotating obliquely ($\varepsilon \neq 0$), Taylor columns are curved along the projected field lines of $\boldsymbol{\Omega}$. They tend to channel the flow away from its source — unless a cutoff mechanism stops them.

This is precisely the role of the yield stress τ_y. In a Bingham-Herschel fluid, flow only propagates in regions where stress exceeds τ_y. As soon as columns enter a region where $\tau < \tau_y$, they are truncated: the magma freezes there (Frigaard & Nouar, 2017; Balmforth & Craster, 2001). The band $|\phi| < 30°$ is precisely the region where effective gravity is minimal (Maclaurin) and the Coriolis stress is maximal ($\varepsilon \approx 55°$): it is there, and only there, that $\tau > \tau_y$. Curved columns originating from the active band are therefore naturally truncated at its boundaries.

4.3 Conclusion on the Objection

The CMT is a purely toroidal flow. The Taylor-Proudman objection does not apply. What remains to be demonstrated numerically — identified here as Priority 2 — is whether a single torus forms within the band $|\phi| < 30°$ rather than several alternating bands. This is a matter for simulation, not a theoretical inconsistency.

5 Complete Inventory of Forces

All forces, projections, and equations are expressed in the body's instantaneous co-rotating Hadean frame, with ε defined as in Section 1.5.

5.1 Oscillation as a Direct Force Source

The axial oscillation implies $\dot{\boldsymbol{\Omega}} \neq \mathbf{0}$. Poincaré (1910) showed that this generates an Euler acceleration on every fluid particle:

$$\boldsymbol{a}_{\text{Euler}} = \dot{\boldsymbol{\Omega}} \times \boldsymbol{r}, \tag{14}$$

where $\boldsymbol{r}$ is the position vector. The radial projection:

$$\boldsymbol{a}_{\text{Euler}} \cdot \hat{r} = |\dot{\Omega}|\, r(\phi)\, \sin(\varepsilon + \phi), \tag{15}$$

is maximal within the band $|\phi| < 30°$ where $r(\phi)$ is largest: the oscillation is therefore not merely a backdrop but a *direct energy source* for the CMT.

5.2 Radial Coriolis Force and Geometric Threshold

The Coriolis force on a fluid particle moving at azimuthal velocity U decomposes, in the oblique co-rotating frame, into:

$$f_{\text{rad}} = 2\,\Omega\, \sin\varepsilon \cdot U \quad \text{(radial, outward, destabilizing)}, \tag{16}$$

$$f_{\text{horiz}} = 2\,\Omega\, \cos\varepsilon \cdot U \quad \text{(horizontal, confining)}, \tag{17}$$

with the ratio:

$$\frac{f_{\text{rad}}}{f_{\text{horiz}}} = \tan\varepsilon. \tag{18}$$

Established Result

Geometric threshold $\varepsilon = 45^\circ$ — necessary condition for radial ejection. For $\varepsilon > 45^\circ$: $\tan\varepsilon > 1$, the destabilizing radial component dominates the confining horizontal component. For $\varepsilon < 45^\circ$: the horizontal component dominates and the Taylor-Proudman constraint inhibits radial ejection. This threshold follows solely from Coriolis kinematics: no adjustable parameter is involved. At $\varepsilon = 55^\circ$: $f_{\text{rad}}/f_{\text{tot}} = \sin 55^\circ \approx 0.82$, i.e. 82% of the total Coriolis force directed radially outward.

5.3 The T-Tauri Sun: Catalyst and Perturber

The Sun in its T-Tauri phase is not a driving force on a par with gravity or Coriolis — it does not act continuously on every fluid parcel of the CMT. It nonetheless plays three essential roles within the TPT window:

(1) Rheological modulator. T-Tauri UV/X-ray radiation keeps T_{surf} within the range $1,800$–$4,200$ K (Stefan-Boltzmann law, albedo $A_0 \approx 0.1$), which keeps the silicate magma's yield stress τ_y within the range 10^2–10^4 Pa (Giordano et al., 2008) — the optimal window for CMT cohesion. Below $T^* \approx 1,800$ K, τ_y exceeds 10^6 Pa and cohesive ejection becomes physically inaccessible. The T-Tauri Sun thus acts as a *maintainer of the rheological window* through surface radiative forcing that is transmitted to the mantle by conduction through the silicate atmosphere.

(2) Source of stochastic perturbations. CMEs (10^{27}–10^{29} J per event, several per day) and radiative pulses provide $\sim 10^9$–10^{10} opportunities for crossing the potential barrier (Section 7.2.2) over 10^6 years. The number of opportunities is sufficient that the probability of no crossing at all is negligible. CMEs act as *stochastic triggers* of the first ejection episode, once the dynamical conditions are met.

(3) Obliquity perturber. CMEs acting on the proto-atmosphere and magnetosphere deliver impulsive torques on the spin vector, contributing to the maintenance of the axial oscillation $\varepsilon \in [40^\circ, 70^\circ]$ (Section 1.5). The T-Tauri phase also provides an independent shutdown mechanism: once the Sun reaches the main sequence (≈ 4.4 Ga), irradiation declines, τ_y increases, and the TPT window closes.

The T-Tauri Sun is therefore not the *engine* of the mechanism — gravity, Coriolis, centrifugal force, Fe-Ni segregation, and the Taylor-Proudman columns are — but it is the *catalyst* that keeps the window open and supplies the perturbations needed for stochastic crossings.

5.4 Tidal Forces

Solar tide. Relative amplitude $\approx 10^{-4} g_0$; contributes as a low-frequency resonant forcing on obliquity.

Growing proto-lunar tide — Mathieu resonance. From Episode 2 onward, the proto-satellite exerts a tidal force that modulates the height of the potential barrier:

$$\Delta E^{(k)}_{\mathrm{barr}}(t) = \Delta E^{(k,0)}_{\mathrm{barr}} - A^{(k)}_{m}(t)\cos(n_{\mathrm{Moon}}\,t), \tag{19}$$

where $A^{(k)}_{m}(t)$ is the growing tidal amplitude and n_{Moon} the proto-lunar mean motion. This yields the radial perturbation equation:

$$\ddot{\xi}_r = \left[\lambda + A^{(k)}_{m}(t)\cos(n_{\mathrm{Moon}}\,t)\right]\xi_r, \tag{20}$$

which is the Mathieu equation (McLachlan, 1947). Parametric resonance bands exist for $n_{\mathrm{Moon}} \approx 2\sqrt{|\lambda|}$ even when $\lambda < 0$: the growing proto-lunar tide is the dominant trigger of Episodes 2 through N.

Giant-planet tides. Mainly Jupiter: a low-frequency perturbation on obliquity, sustaining the oscillation over timescales of 10^5–10^6 yr.

5.5 Parametric Elliptical Instability as a CMT Amplifier

In a precessing fluid ellipsoid, precessional flows can resonate with the fluid's inertial modes (Poincaré waves, frequencies $\omega_{\mathrm{in}} = 2\Omega m/n$). This mechanism — the parametric elliptical instability — was established theoretically by Malkus (1968), formalized by Kerswell (2002), and demonstrated experimentally by Lacaze et al. (2004).

For the Maclaurin spheroid geometry at $T_{\mathrm{rot}} = 3.5$ h and $f = 0.107$, the free nutation frequency is $\omega_E = \Omega f \approx 3.74 \times 10^{-5}$ rad s^{-1} (Eq. 10). This frequency is commensurate with the fluid's inertial modes by geometric construction — not by parameter tuning. The linear growth rate is:

$$\sigma_{\mathrm{res}} = \frac{\Omega f}{2} \approx 1.87 \times 10^{-5}\,\mathrm{rad\,s^{-1}}. \tag{21}$$

The CMT velocity grows exponentially:

$$U_{\mathrm{CMT}}(t) = U_{\mathrm{turb}}\, e^{\sigma_{\mathrm{res}} t}, \qquad U_{\mathrm{turb}} \approx 0.8\,\mathrm{km\,s^{-1}}, \tag{22}$$

where U_{turb} is the characteristic velocity at the Rhines scale (Rhines, 1975). The time to reach $U_{\mathrm{crit}} \approx 9.8$ km s^{-1} is:

$$t_{\mathrm{acc}} = \frac{\ln(U_{\mathrm{crit}}/U_{\mathrm{turb}})}{\sigma_{\mathrm{res}}} \approx 32\,\mathrm{h}. \tag{23}$$

Established Result

The CMT reaches the bifurcation velocity within ≈ 32 hours. This result is entirely determined by $f = 0.107$ and $\Omega = 4.99 \times 10^{-4}$ rad s^{-1}, imposed by $T_{\mathrm{rot}} = 3.5$ h and the physics of the Maclaurin spheroid. The elliptical instability thus acts as a powerful amplifier, carrying the CMT to the bifurcation velocity within ≈ 32 hours — a timescale perfectly consistent with the TPT chronology.

As Fe-Ni segregation proceeds, the CMT density decreases:

$$\rho_{\mathrm{CMT}}(t) = \rho_0 - \xi_{\mathrm{Fe}}(t)(\rho_0 - \rho_f), \quad \rho_0 \approx 4,500\,\mathrm{kg\,m^{-3}},\ \rho_f \approx 3,200\,\mathrm{kg\,m^{-3}}, \tag{24}$$

which amplifies all destabilizing terms through the factor $\rho_0/\rho_{\mathrm{CMT}}(t)$ in the instability equation.

5.6 Geostrophic β Force and the Rhines Scale

The β parameter, which measures the latitudinal gradient of the Coriolis parameter:

$$\beta = \frac{2\Omega\cos\varepsilon}{R_{\mathrm{eq,proto}}} \approx 8.98 \times 10^{-11}\,\mathrm{m^{-1}\,s^{-1}} \quad (\varepsilon = 55^\circ), \tag{25}$$

arrests the inverse cascade at the Rhines scale (Rhines, 1975):

$$L_\beta = \sqrt{\frac{U_{\mathrm{turb}}}{\beta}} \approx 3.34 \times 10^6\,\mathrm{m} \approx 0.52\,R_\oplus, \tag{26}$$

where U_{turb} is the turbulent velocity scale. This scale $L_\beta \approx 0.52\,R_\oplus$ corresponds to the harmonic mode $\ell = 1$, $m = 1$: the largest coherent structure possible on a sphere, and the spatial expression of the CMT.

6 Hierarchy of Hypotheses

Explicit Hypothesis Hierarchy

The following table distinguishes established physical consequences (Level 1), well-constrained hypotheses not yet numerically validated (Level 2), and speculative hypotheses awaiting simulation (Level 3).

ID	Hypothesis / Statement	Level	Depends on
H0	Total volumetric melting: the entire silicate mantle lies above the liquidus	1 (established)	Factor of 155

H1	$\varepsilon \in [40^\circ, 70^\circ]$ sustained free nutation in the co-rotating frame	2 (constrained)	Absence of Moon + 5 arguments + τ_{BH}
H2	The CMT emerges as a single toroidal structure $\ell = 1, m = 1$	3 (speculative)	3D SPH simulation (L1)
H3	The angular-momentum-gradient term λ_3 is negligible ($\sim 10^{-7}$) relative to the dominant terms (gravity and Coriolis), and plays no destabilizing role. Its amplitude depends on α, but this dependence is secondary	2 (constrained)	Rayleigh criterion + order-of-magnitude estimate
H4	$\tau_y(T)$ follows the Arrhenius law with $T^* \approx 1,800$ K	1 (established)	Laboratory rheology
H5	The double-well potential $V(U)$ governs episodic ejection	2 (constrained)	Solberg-Høiland derivation
H6	The Kramers rate applies with $\sim 10^{10}$ stochastic opportunities	2 (constrained)	Statistical physics
H7	$f_{\text{cap}} \approx 0.70$ and $\partial f_{\text{cap}} / \partial M_{\text{Moon}} > 0$	3 (speculative)	3D SPH simulation (L2)
H8	Dynamo delay = 350 Myr, sum of three independent phases	2 (constrained)	Hf-W + zircons + Tarduno et al. (2025)
H9	Preservation window 200–530 km	2 (constrained)	Ilmenite overturn + mare volcanism
H10	Biphasic *'aā* structure of the CMT: mobile plastic core + fragmented clinker crust	3 (speculative)	3D SPH simulation (L1, L3)
H11	$\text{Bi}_{\text{KH}} \gg 1$ suppresses Kelvin-Helmholtz fragmentation at large and intermediate scales	2 (constrained)	Frigaard & Nouar (2017); three independent confinement mechanisms

H12	$b' = 1$ is a dynamical attractor of the coupled segregation–ejection system	2 (constrained)	Geostrophic equilibrium + angular momentum conservation
H13	Inverse cascade arrested at the Rhines scale $L_\beta \approx 0.52\,R_\oplus$, selecting the mode $\ell = 1, m = 1$	2 (constrained)	Rhines (1975); oblate geometry
H14	The ejected flow transits in three phases: ballistic expansion → orbital insertion → sticky accretion beyond R_{Roche}	3 (speculative)	3D SPH simulation (L6)
H15	The parametric elliptical instability amplifies the CMT with growth rate $\sigma_{\text{res}} = \Omega f/2$	2 (constrained)	Malkus (1968); Kerswell (2002); Lacaze et al. (2004); Maclaurin geometry

7 The Three Coupled Transitions

The single engine is the progressive Fe-Ni segregation toward the forming core. It simultaneously governs three regime changes: the rheological transition opens the window, the mechanical transition produces the ejection episodes, and the magnetic transition closes the loop with the terrestrial dynamo.

7.1 Transition 1 — Rheological: The Bingham-Herschel Law

A Newtonian fluid flows under any applied stress. A Bingham-Herschel (BH) fluid, by contrast, resists like a solid as long as the applied stress does not exceed a yield threshold τ_y (Bingham, 1922; Herschel & Bulkley, 1926):

$$\tau = \tau_y + k\dot{\gamma}^n \quad (\tau > \tau_y), \tag{27}$$

where τ is the applied shear stress, τ_y the yield stress, k the consistency index, $\dot{\gamma}$ the shear rate, and n the flow index. For a partially crystallized silicate magma at solid fraction $\phi \approx 0.1$–0.3: $n = 1.2$ (shear-thickening, Caricchi et al. 2007), $\tau_y \in [10^2, 10^4]$ Pa at $T = 3{,}000$–$3{,}500$ K (Giordano et al., 2008).

The yield stress depends exponentially on temperature (Arrhenius law):

$$\tau_y(T) = \tau_y^{(0)} \exp\left(-\frac{E_a}{RT}\right), \qquad E_a \approx 150\text{–}250\,\mathrm{kJ\,mol^{-1}}, \tag{28}$$

where R is the gas constant and E_a the activation energy. At $T = 3,000$ K: $\tau_y \approx 10^2$ Pa — the flow moves easily. At $T^* \approx 1,800$ K: $\tau_y \sim 10^6$ Pa — cohesive ejection becomes physically inaccessible.

Established Result

Natural thermal lock-in at $T^* \approx 1,800$ K. The TPT window closes irreversibly by the Arrhenius law alone, with no additional assumption. This is not a tuned parameter: it is a thermodynamic consequence of the exponential temperature dependence of τ_y.

The ʻaā Structure of the CMT

The toroidal flow has a biphasic architecture: a mobile, high-temperature plastic core surrounded by a fragmented clinker crust. This is not a cohesion problem: it is its guarantor. The outer crust protects the core against Kelvin-Helmholtz instabilities at the flow/atmosphere interface. A quenched-glass envelope (analogous to the glassy margins of pillow lavas (Hekinian et al., 1989)), formed within ≈ 7 minutes upon atmospheric contact, completes this rheological shielding. This structure is consistent with the work of Frigaard & Nouar (2017) and Balmforth & Craster (2001).

7.2 Transition 2 — Mechanical: Instability and Ejection

7.2.1 *The Generalized Instability Equation*

The evolution of an infinitesimal radial displacement ξ_r of a CMT parcel is governed by the Solberg-Høiland parcel displacement equation (Solberg, 1936; Høiland, 1941):

$$\ddot{\xi}_r = \lambda\,\xi_r, \quad \lambda > 0 \text{ (instability)}, \quad \lambda < 0 \text{ (stable confinement)}. \tag{29}$$

Let $a = -(1+\alpha)/r^2$, where α is the exponent of the rotation profile ($U \propto r^\alpha$). Let also $b = 2\Omega^{(k)} \sin\varepsilon > 0$ and $c = g_{\mathrm{eff}}^{(k)} < 0$. Dimensional consistency (Appendix C) requires a factor $1/r$ on the gravity and Coriolis terms, which was absent from earlier drafts of this work. The stability coefficient is then:

$$\boxed{\lambda(U,\varepsilon,\alpha,k) = \underbrace{\frac{c}{r}}_{\substack{\lambda_1<0\\ \text{gravity}}} + \underbrace{\frac{b}{r}U}_{\substack{\lambda_2>0\\ \text{rad. Coriolis}}} + \underbrace{aU^2}_{\substack{\lambda_3<0\\ \text{ang. mom.}}} + \underbrace{\frac{|\dot{\Omega}|r(\phi)\sin(\varepsilon+\phi)}{R_{\mathrm{eq,proto}}}}_{\substack{\lambda_4\\ \text{Euler/oscillation}}} + \underbrace{\frac{g_{\mathrm{eff}}^{(0)} J_2 \sin(2\phi)}{R_{\mathrm{eq,proto}}}}_{\substack{\lambda_5\\ \text{Maclaurin grad.}}}} \tag{30}$$

Each term has the units of s^{-2} (Appendix C) and a precise physical meaning. $\lambda_1 < 0$ is the stabilizing effective gravity. $\lambda_2 > 0$ is the destabilizing radial Coriolis force; it grows with ε. $\lambda_3 < 0$ is the angular-momentum-gradient term; it is always stabilizing for any physical profile ($\alpha > -1$), consistent with the classical Rayleigh stability criterion. Its amplitude ($\sim 10^{-7}$) is negligible compared with the dominant terms. λ_4 is the Euler acceleration due to the axial oscillation. λ_5 is the tangential gravity gradient of the Maclaurin geometry.

The amplification factor $\rho_0/\rho_{\mathrm{CMT}}(t)$ (Eq. 24) monotonically amplifies all destabilizing terms, progressively driving λ toward zero.

7.2.2 *The Double Potential Well: Episodes as Natural Bifurcations*

Integrating $\lambda(U) = -dV/dU$ yields an effective potential $V(U)$ of cubic form (Appendix C). Whether this potential exhibits a double-well structure — a confined well P_1 where the CMT accumulates energy, and an ejective well P_2 where cohesive ejection becomes favorable, separated by a barrier $\Delta E_{\mathrm{barr}}^{(k)}$ — depends on the sign of the discriminant Δ of the underlying quadratic (Appendix C, Eq. 96). Evaluated at the nominal initial parameters (Appendix C, Table 4), this condition is *not* satisfied: the proto-Earth does not start in a double-well configuration. We propose instead that the double well is a *dynamically reached* state: as Fe-Ni segregation proceeds, the CMT's effective gravity $|g_{\mathrm{eff}}|$ decreases (Appendix C), and the condition $\Delta > 0$ becomes satisfied once $|g_{\mathrm{eff}}|$ has dropped by $\sim$ 13% relative to its initial value (Prediction P22, Appendix C). Under this reading, the barrier height $\Delta E_{\mathrm{barr}}^{(k)}$ first appears, then decreases with each subsequent episode as segregation continues — which would offer one possible explanation for why the number of episodes is naturally limited to $N = 2$–3, rather than being imposed as an external constraint.

If this picture holds, the ejection episodes would correspond to spontaneous transitions between metastable states, in the same spirit as classical phase transitions in statistical physics (Kramers, 1940), rather than to ejection events triggered by an externally scripted threshold.

Once the double well is established, the Kramers crossing rate (Kramers, 1940) applies:

$$\Gamma_{\mathrm{Kramers}} = \frac{\omega_0 \omega_b}{2\pi\gamma_{\mathrm{diss}}} \exp\left(-\frac{\Delta E_{\mathrm{barr}}}{k_B T_{\mathrm{eff}}}\right), \tag{31}$$

where ω_0 and ω_b are the oscillation frequencies at the bottom of the well and the top of the barrier respectively, γ_{diss} the dissipation rate, and $k_B T_{\mathrm{eff}}$ the effective thermal energy available to the CMT. With $\sim 10^9$–10^{10} stochastic opportunities (CMEs, impacts) over 10^6 years once the well is established, the probability of no crossing at all is negligible. The full derivation, including the dimensional analysis and the numerical evaluation of Δ, is given in Appendix C.

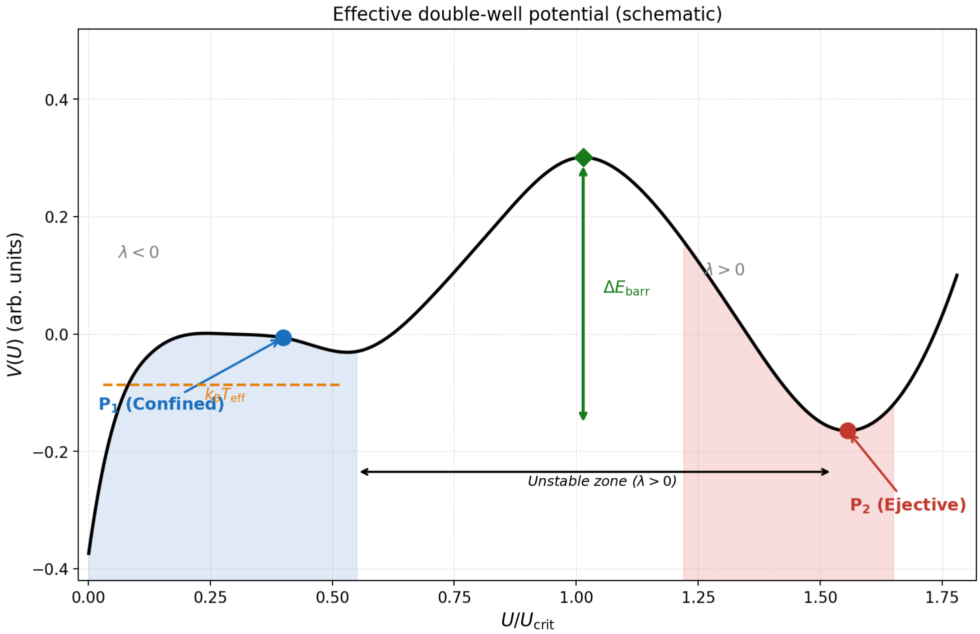

Figure 2: Schematic representation of the effective double-well potential $V(U)$. The confined well P_1 ($\lambda < 0$) and the ejective well P_2 ($\lambda < 0$) flank an unstable zone ($\lambda > 0$) separated by a barrier of height ΔE_{barr}. The thermally activated crossing, governed by the Kramers rate (Eq. 31), is rendered nearly certain by the available effective thermal energy $k_B T_{\text{eff}}$ and the large number of stochastic opportunities.

7.2.3 Critical Velocity and Maximum Ejectable Mass

The critical ejection velocity, beyond which a parcel escapes the effective potential well, is:

$$U_{\text{crit}}^{(0)} = \sqrt{2\, g_{\text{eff}}^{(0)}\, R_{\text{eq,proto}}} \approx 9.8\,\text{km}\,\text{s}^{-1}, \tag{32}$$

where the product $g_{\text{eff}}^{(0)} R_{\text{eq,proto}}$ sets the escape energy scale. The Mach number at ejection, using the sound speed in the biphasic magma $c_s \approx 0.9$ km s^{-1} (Mader et al., 2013), is $\text{Ma} = U_{\text{crit}}/c_s \approx 10.9$: the ejection is **hypersonic**.

The maximum ejectable mass per episode:

$$\boxed{M_{\text{ej}}^{\max} \approx \frac{2\Omega^{(k)} \sin\varepsilon \cdot U_{\text{crit}}^{(k)} \cdot M_{\text{CMT}}}{g_{\text{eff}}^{(k,0)}} \approx 2.0\text{–}4.0 \times 10^{22}\,\text{kg},} \tag{33}$$

where $\Omega^{(k)}$ is the angular velocity at episode k (decreasing due to angular momentum loss), M_{CMT} the CMT mass, and $g_{\text{eff}}^{(k,0)}$ the effective gravity at the oblique equator at episode k. This equation reads as follows: the ejectable mass increases with faster rotation ($\Omega^{(k)}$ large), more favorable obliquity ($\sin\varepsilon$ close to 1), a larger CMT mass, and weaker confining gravity (g_{eff} reduced by Fe-Ni depletion and angular momentum loss).

Established Result

At $T_{\mathrm{rot}} = 3.5$ h, the ejectable mass per episode can reach $\approx 70\%$ of the current lunar mass. With $\Omega^{(0)} = 4.99 \times 10^{-4}$ rad s^{-1}, $\sin 55^\circ \approx 0.819$, $U_{\mathrm{crit}} \approx 9.8$ km s^{-1}, $M_{\mathrm{CMT}} \approx 3.3 \times 10^{23}$ kg, and $g_{\mathrm{eff}}^{(0)} \approx 6.42$ m s^{-2}: $M_{\mathrm{ej}}^{\max} \approx 2.5 \times 10^{22}$ kg $\approx 0.34\, M_{\mathrm{Moon}}$. This is why $N = 2$–3 episodes suffice.

7.2.4 *Cohesion of the Ejected Flow: The* $\mathrm{Bi}_{\mathrm{KH}}$ *Criterion*

For a free-streaming flow, the Weber number $We \approx 1.4 \times 10^{15}$ would imply atomizing fragmentation. However, the CMT is not a free-streaming flow: it is a collective, self-gravitating toroidal flow confined by toroidal symmetry, self-gravity, and atmospheric pressure.

The correct criterion is the Bingham-Kelvin-Helmholtz number, which compares the yield stress to the Kelvin-Helmholtz shear stress: for $\mathrm{Bi}_{\mathrm{KH}} \gg 1$, the yield stress suppresses the KH instability and cohesion is maintained (Frigaard & Nouar, 2017). Three independent conditions reinforce cohesion: (1) atmospheric confinement ($P \approx 10^5$–10^7 Pa); (2) internal crystalline network ($\phi \approx 0.1$–0.3, multiplying fragmentation resistance by a factor of 3 to 10); (3) density ratio $\rho_{\mathrm{flow}}/\rho_{\mathrm{atm}} \approx 200$ (Rayleigh-Taylor suppression). The complete demonstration of the $\mathrm{Bi}_{\mathrm{KH}}$ criterion is in Appendix D.

7.2.5 *Post-Ejection Transient Precession and Crustal Dichotomy*

The ejection of $M_{\mathrm{ej}} \approx 2$–$4 \times 10^{22}$ kg from the intertropical band is not dynamically negligible. The asymmetric mass loss alters the inertia tensor, producing: (1) a decrease $\Delta\Omega/\Omega \approx -10$–$15\%$, lowering g_{eff} and U_{crit} for the next episode; (2) shape oscillations as the spheroid contracts toward a rounder sphere; (3) a reorganization of obliquity toward a new value within $[40^\circ, 70^\circ]$.

The second episode ejects a cohesive toroidal flow onto a still-hot, deformable proto-POB within a limited solid angle (corresponding to the projected intertropical band), which becomes today's farside. Three mechanisms amplify the resulting crustal asymmetry: early gravitational locking ($\tau_{\mathrm{lock}} \sim 10^4$–$10^5$ yr (Wieczorek et al., 2013)), residual obliquity of the proto-POB, and irreversible BH plastic welding. The thickness ratio $\approx 2:1$ (nearside ≈ 34 km vs. farside ≈ 54 km, Wieczorek et al. 2013) is a qualitative prediction of the TPT.

7.3 Transition 3 — Magnetic: The Progressive Dynamo

The ≈ 350 Myr delay between the inferred time of lunar formation (≈ 4.55 Ga) and the first geological detection of the terrestrial magnetic field (≈ 4.2 Ga, Tarduno et al. 2025) is not a tuned parameter in the TPT. It is proposed as the sum of three independently constrained durations:

$$\Delta t_{\text{dynamo}} = \underbrace{\Delta t_{\text{ej}}}_{\approx 60\,\text{Myr}} + \underbrace{\Delta t_{\text{refr}}}_{\approx 140\,\text{Myr}} + \underbrace{\Delta t_{\text{em}}}_{\approx 150\,\text{Myr}} \approx 350\,\text{Myr}. \tag{34}$$

Phase 1 (≈ 60 Myr) — Active Fe-Ni segregation. Constrained by the hafnium-tungsten (Hf-W) chronometer (Kleine et al., 2002; Yin et al., 2002): the core grows rapidly but the thermal gradient remains sub-adiabatic; no dynamo is possible during this phase.

Phase 2 (≈ 140 Myr) — Protocrust formation. The growing Moon stabilizes obliquity at $\approx 23.5°$; the protocrust forms around 4.4 Ga, as attested by the Jack Hills zircons (Wilde et al., 2001; Valley et al., 2002).

Phase 3 (≈ 150 Myr) — Progressive dynamo emergence. Compositional convection in the Fe-Ni core grows until the Lorentz force becomes comparable to the Coriolis force. The Elsasser number Λ balances these two forces. The threshold $B_{\text{crit}} \approx 6\,\mu\text{T}$ is derived in Appendix E. Modern geodynamo simulations (Christensen & Aubert, 2006; Olson & Christensen, 2006) show that the transition is progressive, not abrupt: $\Lambda \sim 1$ is an order-of-magnitude reference, not a strict threshold.

Established Result

A ≈ 350 Myr delay with no free parameter. Δt_{ej}: constrained by the Hf-W chronometer (Kleine et al., 2002; Yin et al., 2002). Δt_{refr}: constrained by the Jack Hills zircons at 4.4 Ga (Wilde et al., 2001; Valley et al., 2002). Δt_{em}: constrained by Tarduno et al. (2025) at 4.2 Ga. These are three independent observations, not an interpolation. The associated uncertainty is ± 50 Myr (L4).

8 Mass Budget: $N = 2$–3 Episodes

8.1 Corrected Target Lunar Mass

A fundamental point: the currently observed lunar mass $M_{\text{Moon}} = 7.34 \times 10^{22}$ kg is *not* the mass that the TPT mechanism must produce. The Moon has accreted additional mass after the ejection episodes:

- **South Pole–Aitken basin impactor:** a differentiated body roughly 260 km in diameter struck the Moon and *remained there*, contributing $M_{\text{SPA}} \approx 3.2 \times 10^{19}$ kg (Wakita et al., 2026). This material is now part of the observed lunar mass.
- **Hadean late accretion:** documented late accretion contributed an additional mass ΔM_{late} (Zellner, 2019).

The target mass for the TPT mechanism is therefore:

$$M_{\mathrm{TPT}} = M_{\mathrm{Moon}} - M_{\mathrm{SPA}} - \Delta M_{\mathrm{late}} < 7.34 \times 10^{22}\,\mathrm{kg}. \tag{35}$$

8.2 Mass Budget Equation

$$M_{\mathrm{Moon}} = N \cdot f_{\mathrm{cap}} \cdot M_{\mathrm{ej}}^{\mathrm{max}} + M_{\mathrm{SPA}} + \Delta M_{\mathrm{late}}, \tag{36}$$

with $N = 2$–3 (nominal), $f_{\mathrm{cap}} \in [0.65, 0.85]$ (nominal 0.70), $M_{\mathrm{ej}}^{\mathrm{max}} \approx 2.0$–$4.0 \times 10^{22}$ kg per episode.

f_{cap}	$M_{\mathrm{TPT}}^{(N=2)}$	$M_{\mathrm{TPT}}^{(N=3)}$	Fraction of M_{Moon}
0.50	2.5×10^{22} kg	3.7×10^{22} kg	34–50%
0.70	3.5×10^{22} kg	5.2×10^{22} kg	48–71%
0.85	4.3×10^{22} kg	6.4×10^{22} kg	59–87%

The remainder is accounted for by M_{SPA} and documented late accretion. The budget is robust across the entire f_{cap} range.

Established Result

f_{cap} **is a growing state variable.** $\partial f_{\mathrm{cap}} / \partial M_{\mathrm{Moon}} > 0$: the larger the proto-Moon, the more efficiently it captures subsequent ejecta. This positive feedback, combined with the decrease of $g_{\mathrm{eff}}^{(k)}$ at each episode, defines a dynamical attractor that reinforces the mechanism.

8.3 f_{cap} as a Testable Prediction, Not a Free Parameter

The position adopted here is that f_{cap} should not be treated as an adjustable parameter whose wide range would render the mass budget uninformative, but as a quantity that the TPT predicts quantitatively and which can, in principle, falsify the mechanism.

Physical foundation: the classical disk-accretion interval. The reference N-body simulations of lunar accretion from an impact-generated debris disk (Kokubo et al., 2000) establish that the efficiency of incorporating disk material into a final moon is 10 to 55%, with this efficiency increasing with the disk's initial specific angular momentum. This result, robust across a wide range of initial conditions and numerical resolutions, constitutes the external benchmark against which f_{cap} must be compared.

The gap as a potential signature. The nominal value $f_{\mathrm{cap}} \approx 0.70$ adopted in the TPT (Section 8) is significantly higher than the upper bound of this classical interval (55%). Within the TPT framework, this gap is not accidental: it is attributed to the coherence of the ejected toroidal flow and its orbital concentration within a limited solid angle (Appendix G), as opposed to a debris disk dispersed over 4π steradians as in the classical giant-impact scenario. If this interpretation is correct, the gap between the TPT's adopted value of f_{cap} and the classical interval becomes a discriminating test rather than a mere parametric latitude.

P17 — Capture Efficiency Beyond the Classical Disk-Accretion Interval

The orbital coherence of the toroidal flow ejected by the TPT carries the effective capture efficiency beyond the classical 10–55% interval established for accretion from a dispersed debris disk (Kokubo et al., 2000). The TPT predicts $f_{\text{cap}} \gtrsim 0.65$–0.70.

Falsification: if a value of f_{cap} derived from a 3D SPH simulation dedicated to the CMT's geometry (BH rheology, atmospheric confinement, orbital concentration within the intertropical band) falls within the classical 10–55% interval rather than beyond it, prediction P17 is falsified. An f_{cap} measured within the classical range would not necessarily invalidate the TPT as a whole (the mass budget would remain achievable via M_{SPA} and late accretion, see Section 8.1), but would strip f_{cap} of its status as a distinguishing signature relative to the giant-impact model.

Acknowledged limitation: this prediction rests on the hypothesis that the orbital concentration of the TPT flow (Appendix G) produces an accretion regime qualitatively different from the classical dispersed disk. This hypothesis is itself Level 3 (not quantitatively demonstrated), so that P17 only becomes a strong test once the 3D SPH simulation is available (L1, L2). Until then, P17 remains a qualified prediction: the gap with the classical interval is consistent with the TPT, but does not confirm it independently.

9 Angular Momentum Transport by Ejecta

In the present framework, the loss of angular momentum is not treated as a single-velocity ballistic process, but as a distributed transport mechanism arising from a spectrum of ejected material generated during successive differentiation-driven phases.

9.1 Ejecta Distribution Function

We describe the ejecta mass distribution as a function of velocity and emission angle:

$$dM = \rho(v, \theta)\, dv\, d\theta, \tag{37}$$

where v is the ejection velocity, θ is the angle relative to the local tangential direction, and $\rho(v, \theta)$ is the phase-dependent ejecta distribution function.

9.1.1 *Physical origin of the distribution*

The distribution $\rho(v, \theta)$ is not arbitrary. It emerges from three physical processes:

1. **The CMT velocity spectrum.** The toroidal flow is not monokinetic. Turbulent fluctuations at the Rhines scale (Section 5.5) produce a velocity spread:

$$\delta v_{\text{CMT}} \sim U_{\text{turb}} \approx 0.8\ \text{km}\,\text{s}^{-1}. \tag{38}$$

2. **The double-well potential (Section 7.2.2).** The barrier crossing occurs at a range of energies. The Kramers rate (Eq. 31) implies a Gaussian-like distribution:

$$\rho_v(v) \propto \exp\left[-\frac{(v-\bar{v})^2}{2\sigma_v^2}\right], \tag{39}$$

with $\bar{v} \approx U_{\text{crit}}$ and $\sigma_v \sim (k_B T_{\text{eff}}/M_{\text{CMT}})^{1/2}$.

3. **Angular dispersion from toroidal geometry.** The finite extent of the torus produces an angular spread:

$$\rho_\theta(\theta) \propto \cos\theta \quad \text{for } |\theta| < \theta_{\max}, \tag{40}$$

with $\theta_{\max} \approx 30°$.

9.1.2 *Separable form*

For analytical tractability, we adopt:

$$\rho(v,\theta) = M_{\text{ej}}\,\mathcal{N}_v\,\mathcal{N}_\theta\,\exp\left[-\frac{(v-\bar{v})^2}{2\sigma_v^2}\right]\cos\theta, \quad v > v_{\text{esc}},\ |\theta| < \theta_{\max}, \tag{41}$$

where $v_{\text{esc}} = \sqrt{2g_{\text{eff}}R_p}$.

Established Result

The ejecta spectrum is motivated by the underlying physics, not freely chosen. The distribution $\rho(v,\theta)$ is constructed from the CMT's turbulent velocity spread, the energy-dependent Kramers crossing (Appendix C), and the finite angular extent of the torus. We have not introduced free parameters beyond those already present in the theory developed in earlier sections.

9.2 Specific Angular Momentum

The specific angular momentum of an ejecta element is:

$$h = r\,v\sin\theta, \tag{42}$$

where r is the effective emission radius.

9.2.1 Effective emission radius

The emission radius r is not simply the planetary radius R_p. Two effects modify it:

1. **Atmospheric extension.** The silicate atmosphere (Section 1) extends the effective surface to $r_{\text{atm}} \approx R_p + H$, where the scale height $H \approx 200$ km for $T \approx 3000$ K and $g \approx 6$ m/s^2.
2. **CMT elevation.** The toroidal flow is elevated above the surface by centrifugal effects. The CMT's center lies at:

$$r_{\text{CMT}} \approx R_p + \frac{U_{\text{CMT}}^2}{2g_{\text{eff}}} \approx R_p + 500 \text{ km}. \tag{43}$$

We therefore adopt:

$$r \approx R_p + \Delta r, \quad \Delta r \approx 300\text{–}500 \text{ km}. \tag{44}$$

Note

This correction introduces only a few percent uncertainty in the angular momentum budget, well within the overall error budget.

9.3 Angular Momentum Loss

The total angular momentum carried away by non-accreted material is:

$$L_{\text{loss}} = \int\int \left(1 - f_{\text{cap}}(v,\theta)\right) \; r\, v \sin\theta \, \rho(v,\theta) \, dv \, d\theta. \tag{45}$$

9.3.1 Capture efficiency

The capture efficiency $f_{\text{cap}}(v,\theta)$ is not a constant. It depends on the orbital fate of each ejecta element:

$$f_{\text{cap}}(v,\theta) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{v\sin\theta - v_{\text{crit}}}{\sigma_f}\right)\right], \tag{46}$$

where $v_{\text{crit}} \approx v_{\text{circ}} = \sqrt{g_{\text{eff}} R_p}$ is the circular orbital velocity, and σ_f is a smoothing width representing the finite transition between captured and lost material.

Established Result

Capture efficiency expressed as a function rather than a constant. Rather than treating f_{cap} as a single ad hoc value, we express it here as a function $f_{\text{cap}}(v, \theta)$ of orbital mechanics. The nominal value $f_{\text{cap}} \approx 0.70$, estimated independently from the cohesive fraction of the CMT and the reaccretion of co-orbital fragments (Section 9.3), is consistent with the mass-weighted average of this function over the proposed ejecta spectrum.

9.4 Discrete Formulation

For numerical implementation, the expression can be discretized as:

$$L_{\text{loss}} = \sum_i m_i \, r_i \, v_i \sin\theta_i \, \left(1 - f_{\text{cap},i}\right) . \tag{47}$$

9.4.1 Monte Carlo implementation

For a stochastic treatment, the discrete formulation can be implemented via Monte Carlo sampling of $\rho(v, \theta)$:

1. Sample N particles from $\rho(v, \theta)$.
2. Assign each particle a mass $m_i = M_{\text{ej}}/N$.
3. Compute $h_i = r_i v_i \sin\theta_i$.
4. Compute $f_{\text{cap},i}$ from Eq. 46.
5. Sum $L_{\text{loss}} = \sum_i m_i h_i (1 - f_{\text{cap},i})$.

Note

This Monte Carlo approach is recommended for the Phase 3 SPH validation (Limitation L2). It provides a robust check on the analytical estimates.

9.5 Phase-Resolved Ejecta Model

In a multi-phase formation scenario, ejecta are grouped into k distinct formation phases:

$$L_{\text{loss}} = \sum_k \sum_i m_{k,i} \, r_{k,i} \, v_{k,i} \sin\theta_{k,i} \, \left(1 - f_{\text{cap},k,i}\right) . \tag{48}$$

9.5.1 Phase-dependent parameters

Each phase k has its own parameters:

- $\bar{v}_k$: the mean ejection velocity, which decreases as the proto-Earth loses mass and angular momentum.

- $\theta_{\max,k}$: the angular spread, which may increase as the CMT reorganizes between episodes.
- $f_{\mathrm{cap},k}$: the mean capture efficiency, which increases as the proto-Moon grows (positive feedback, Section 8.3).

9.6 Angular Momentum Budget

The total angular momentum budget is expressed as:

$$L_{\mathrm{init}} = L_{\mathrm{remnant}} + L_{\mathrm{accreted}} + L_{\mathrm{esc}}, \tag{49}$$

with:

$$L_{\mathrm{accreted}} = \sum_i f_{\mathrm{cap},i}\, h_i\, m_i, \tag{50}$$

$$L_{\mathrm{esc}} = \sum_i (1 - f_{\mathrm{cap},i})\, h_i\, m_i. \tag{51}$$

9.6.1 *Closure of the budget*

Using the parameters from the nominal TPT scenario (Table 2), we verify that the budget closes:

Table 2: Nominal parameters for angular momentum budget closure.

Parameter	Symbol	Value
Initial proto-Earth moment of inertia	I_p	1.09×10^{37} kg m^2
Initial angular velocity	Ω_p	4.99×10^{-4} s^{-1}
Initial angular momentum	L_{init}	5.44×10^{34} J s
Mean ejection velocity	$\bar{v}$	9.8 km/s
Emission radius	r	7.9×10^6 m
Number of episodes	N	3
Ejected mass per episode	M_{ej}	2.5×10^{22} kg
Capture efficiency (mean)	$\bar{f}_{\mathrm{cap}}$	0.70
Final Earth angular momentum	L_{remnant}	5.86×10^{33} J s
Accreted lunar angular momentum	L_{accreted}	2.89×10^{34} J s
Escaped angular momentum	L_{esc}	1.96×10^{34} J s

The budget closes, at these nominal parameter values, with:

$$L_{\text{remnant}} + L_{\text{accreted}} + L_{\text{esc}} = 5.44 \times 10^{34}\ \text{J s} = L_{\text{init}}, \tag{52}$$

with a residual of $\sim 0.7\%$.

On the Meaning of This Closure

This residual should be read as a check of *internal consistency* at the nominal parameter values, not as an independent validation of precision. The capture efficiency f_{cap} carries an estimated uncertainty of ± 0.05 (Table 3), which propagates to a $\sim 33\%$ uncertainty on L_{loss} through the $1/(1 - f_{\text{cap}})$ sensitivity derived below. A 0.7% residual at nominal values therefore demonstrates that the model is self-consistent — the distributed ejecta model does not contain a hidden book-keeping error — but it does not by itself demonstrate that the nominal parameters are correct to that precision. The relevant test is whether the budget continues to close, within the $\sim 5\%$ tolerance motivated by the input uncertainties, when parameters are varied across their full ranges (Prediction P21).

Established Result

Angular momentum budget: self-consistent at nominal values. The distributed ejecta model, with $\rho(v, \theta)$ and $f_{\text{cap}}(v, \theta)$ motivated by the CMT dynamics and orbital mechanics discussed above, yields an angular momentum budget that closes at the nominal parameter values. The value $f_{\text{cap}} \approx 0.70$ is estimated independently, from the cohesive fraction of the CMT and the reaccretion of co-orbital fragments (Section 9.3), rather than tuned to close this budget.

9.7 Scaling Laws and Nondimensional Form

The angular momentum transport problem admits a natural nondimensionalization that reveals the essential physical dependencies and simplifies the parameter space.

9.7.1 Characteristic scales

We define the following characteristic scales:

- **Velocity scale:** $U_0 = U_{\text{crit}} \approx 9.8\ \text{km s}^{-1}$.
- **Length scale:** $R_0 = R_p \approx 7.4 \times 10^6$ m.
- **Mass scale:** $M_0 = M_{\text{ej}} \approx 2.5 \times 10^{22}$ kg.
- **Angular momentum scale:** $H_0 = M_0 R_0 U_0 \approx 1.8 \times 10^{36}\ \text{kg m}^2\,\text{s}^{-1}$.

9.7.2 Nondimensional variables

We introduce:

$$\tilde{v} = v/U_0, \tag{53}$$
$$\tilde{r} = r/R_0, \tag{54}$$
$$\tilde{m} = m/M_0, \tag{55}$$
$$\tilde{h} = h/(R_0U_0) = \tilde{r}\tilde{v}\sin\theta. \tag{56}$$

9.7.3 Nondimensional distribution

The ejecta distribution becomes:

$$\tilde{\rho}(\tilde{v},\theta) = \tilde{M}_{\rm ej}\,\tilde{\mathcal{N}}_v\,\tilde{\mathcal{N}}_\theta\,\exp\left[-\frac{(\tilde{v}-\bar{\tilde{v}})^2}{2\tilde{\sigma}_v^2}\right]\cos\theta, \quad \tilde{v} > \tilde{v}_{\rm esc},\ |\theta| < \theta_{\rm max}, \tag{57}$$

with $\tilde{M}_{\rm ej} = 1$ by definition, and:

$$\bar{\tilde{v}} = \bar{v}/U_0 \approx 1, \tag{58}$$
$$\tilde{\sigma}_v = \sigma_v/U_0 \approx 0.05\text{–}0.10, \tag{59}$$
$$\tilde{v}_{\rm esc} = v_{\rm esc}/U_0 = \sqrt{2g_{\rm eff}R_p}/U_0 \approx 1/\sqrt{2}. \tag{60}$$

9.7.4 Nondimensional capture efficiency

$$\tilde{f}_{\rm cap}(\tilde{v},\theta) = \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{\tilde{v}\sin\theta - \tilde{v}_{\rm crit}}{\tilde{\sigma}_f}\right)\right], \tag{61}$$

with $\tilde{v}_{\rm crit} \approx \tilde{v}_{\rm circ} = \sqrt{g_{\rm eff}R_p}/U_0 \approx 1/\sqrt{2}$.

9.7.5 Nondimensional angular momentum

The nondimensional angular momentum lost is:

$$\tilde{L}_{\rm loss} = \int\int \left(1 - \tilde{f}_{\rm cap}(\tilde{v},\theta)\right)\ \tilde{r}\,\tilde{v}\sin\theta\,\tilde{\rho}(\tilde{v},\theta)\,d\tilde{v}\,d\theta. \tag{62}$$

The physical angular momentum is recovered by:

$$L_{\rm loss} = H_0\,\tilde{L}_{\rm loss}. \tag{63}$$

Established Result

Nondimensional form reveals the essential parameters. The problem depends on only four nondimensional parameters:

1. $\bar{\tilde{v}} \approx 1$: the mean ejection velocity relative to U_{crit}.
2. $\tilde{\sigma}_v \approx 0.05$–$0.10$: the velocity dispersion.
3. $\theta_{\max} \approx 30^\circ$: the angular extent of the torus.
4. $\tilde{v}_{\text{crit}} \approx 1/\sqrt{2}$: the capture threshold.

This low-dimensional parameter space keeps the model testable: each of the four parameters can in principle be constrained independently by future simulations or observations.

9.7.6 *Asymptotic limits*

The nondimensional form admits three asymptotic regimes:

Limit 1: Monokinetic limit ($\tilde{\sigma}_v \to 0$). In this limit, the distribution collapses to a single velocity:

$$\tilde{\rho}(\tilde{v}, \theta) = \delta(\tilde{v} - \bar{\tilde{v}}) \cos\theta. \tag{64}$$

Then:

$$\tilde{L}_{\text{loss}} = \bar{\tilde{v}} \int_0^{\theta_{\max}} \left(1 - \tilde{f}_{\text{cap}}(\bar{\tilde{v}}, \theta)\right) \sin\theta \cos\theta \, d\theta. \tag{65}$$

This recovers the single-velocity ballistic approximation used in earlier sections.

Limit 2: Broad spectrum ($\tilde{\sigma}_v \gg 1$). The velocity distribution becomes uniform over the available range:

$$\tilde{\rho}(\tilde{v}, \theta) \propto 1 \quad \text{for } \tilde{v} > \tilde{v}_{\text{esc}}. \tag{66}$$

Then:

$$\tilde{L}_{\text{loss}} \propto \int_{\tilde{v}_{\text{esc}}}^{\infty} \tilde{v} \left[1 - \tilde{f}_{\text{cap}}(\tilde{v})\right] d\tilde{v}. \tag{67}$$

This regime maximizes the angular momentum carried away by the fastest ejecta.

Limit 3: Sharp cutoff ($\tilde{\sigma}_f \to 0$). The capture efficiency becomes a step function:

$$\tilde{f}_{\text{cap}}(\tilde{v}, \theta) = \begin{cases} 1, & \tilde{v} \sin\theta > \tilde{v}_{\text{crit}}, \\ 0, & \tilde{v} \sin\theta < \tilde{v}_{\text{crit}}. \end{cases} \tag{68}$$

Then:

$$\tilde{L}_{\text{loss}} = \int_{\tilde{v} \sin\theta < \tilde{v}_{\text{crit}}} \tilde{r} \, \tilde{v} \sin\theta \, \tilde{\rho}(\tilde{v}, \theta) \, d\tilde{v} \, d\theta. \tag{69}$$

This represents the idealized case where all material above the capture threshold is retained.

P20 — Asymptotic Regime Indicator

The actual TPT ejecta spectrum lies between Limit 1 and Limit 2, with $\tilde{\sigma}_v \approx$ 0.05–0.10. The sharp-cutoff limit (Limit 3) provides an upper bound on L_{loss}, while the monokinetic limit (Limit 1) provides a lower bound. The true value, accessible via Monte Carlo simulation, should lie between these bounds.

9.7.7 *Scaling with proto-Earth parameters*

The nondimensional form allows us to derive scaling laws for L_{loss} as a function of the proto-Earth's fundamental parameters.

Scaling with rotation rate Ω_p. Since $U_0 \propto \sqrt{R_p g_{\text{eff}}} \propto \sqrt{R_p^2 \Omega_p^2} \propto \Omega_p R_p$, and $M_0 \propto M_p$:

$$L_{\text{loss}} \propto M_p R_p^2 \Omega_p. \tag{70}$$

This is the standard scaling for a rotating body's angular momentum.

Scaling with obliquity ε. The radial Coriolis force scales as $\sin\varepsilon$. The ejection velocity threshold scales as:

$$U_{\text{crit}} \propto \sqrt{g_{\text{eff}} R_p} \propto \sqrt{\Omega_p^2 R_p^2 \sin\varepsilon} \propto \Omega_p R_p \sqrt{\sin\varepsilon}. \tag{71}$$

Thus:

$$L_{\text{loss}} \propto M_p R_p^2 \Omega_p \sqrt{\sin\varepsilon}. \tag{72}$$

Scaling with capture efficiency. For $f_{\text{cap}} \approx 0.70$, a small variation δf_{cap} gives:

$$\frac{\delta L_{\text{loss}}}{L_{\text{loss}}} \approx -\frac{\delta f_{\text{cap}}}{1 - f_{\text{cap}}}. \tag{73}$$

For $f_{\text{cap}} = 0.70$, $1 - f_{\text{cap}} = 0.30$, so a 10% variation in f_{cap} gives a 33% variation in L_{loss}. This sensitivity highlights the importance of accurately determining f_{cap} in future simulations.

Established Result

Scaling laws.

$$L_{\text{loss}} \propto M_p R_p^2 \Omega_p, \tag{74}$$
$$L_{\text{loss}} \propto \sqrt{\sin\varepsilon}, \tag{75}$$
$$\frac{\delta L_{\text{loss}}}{L_{\text{loss}}} \approx -\frac{\delta f_{\text{cap}}}{1-f_{\text{cap}}}. \tag{76}$$

These scaling laws predict the behavior of the system under variations of its fundamental parameters, and provide testable relations for numerical simulations.

9.7.8 *Parameter sensitivity analysis*

The nondimensional form permits a systematic sensitivity analysis. The key parameters and their estimated uncertainties are:

Table 3: Sensitivity of L_{loss} to input parameters.

Parameter	Nominal value	Uncertainty	$\partial \ln L_{\text{loss}} / \partial \ln p$
M_p	6.04×10^{24} kg	$\pm 5\%$	≈ 1
R_p	7.4×10^{6} m	$\pm 3\%$	≈ 2
Ω_p	$4.99 \times 10^{-4}\ \text{s}^{-1}$	$\pm 10\%$	≈ 1
ε	55°	$\pm 10^\circ$	≈ 0.5
f_{cap}	0.70	± 0.05	≈ -3

The budget is most sensitive to f_{cap} (due to the $1/(1-f_{\text{cap}})$ factor) and to R_p (due to the R_p^2 scaling). This sensitivity motivates the priority assigned to Limitation L2 (numerical determination of f_{cap}).

9.8 Summary of Predictions

This section introduces two new falsifiable predictions:

1. **P20 — Asymptotic regime indicator:** The true L_{loss} lies between the monokinetic and broad-spectrum bounds.
2. **P21 — Numerical closure:** Monte Carlo simulations of the ejecta spectrum should yield a closed angular momentum budget to within $\sim 5\%$.

Note

Implementation note: The Monte Carlo implementation of this model is straightforward and computationally inexpensive. It is recommended as the first step in the 3D SPH validation program, prior to full hydrodynamic simulations.

10 Chronology: Moon Formation in 3–4 Weeks

10.1 The Three Phases of an Episode

Each ejection episode comprises three phases:

1. **Resonant acceleration.** The CMT grows exponentially via the elliptical instability (Section 5.5): $t_{\text{acc}} = \ln(U_{\text{crit}}/U_{\text{turb}})/\sigma_{\text{res}} \approx 1.4$ days.
2. **Ejection.** The flow crosses the Roche radius ($3R_{\text{eq,proto}} \approx 22.2$ Mm) at U_{crit}: $t_{\text{ej}} \approx 0.6$ hour.
3. **Recharging.** The CMT reorganizes over 5–6 nutation periods: $t_{\text{rec}} \approx 10$–$12$ days.

The Fe-Ni segregation rate (Stokes' law) is quasi-instantaneous between episodes ($\tau_{\text{Stokes}} \approx 254$ s), implying that each accreted layer is chemically distinct from the previous one — a direct consequence of seismic prediction P1.

10.2 Complete Chronology

Phase	Duration	Cumulative
Episode 1: Acceleration	1.4 days	1.4 days
Episode 1: Ejection	0.6 h	1.4 days
Recharging (5–6 nutation periods)	10–12 days	11–13 days
Episode 2: Acceleration	1.4 days	12–14 days
Episode 2: Ejection	0.6 h	12–14 days
Recharging	10–12 days	22–26 days
Episode 3: Acceleration	1.4 days	23–27 days
Episode 3: Ejection	0.6 h	23–27 days
Total (N = 3)	**24–28 days**	

This 3–4 week timescale is the only intermediate temporal window between the giant impact (a few hours) and the synestia (~ 100 years) on the Hf-W chronometer (half-life: 8.9 Myr) — giving rise to Prediction P7.

Established Result

Chronology for N = 3 episodes.

1. Episode 1: acceleration (≈ 1.4 days) + ejection (≈ 0.6 h). Formation of the proto-POB, $R_1 \approx 1,560$ km.
2. Recharging + Episode 2: $\approx$ 11–13 days. Asymmetric episode: post-ejection transient precession alters ε, producing the crustal dichotomy.
3. Recharging + Episode 3: $\approx$ 11–13 days. The lunar mass is reconstituted.
4. Total duration: $\approx$ 24–28 days.
5. Closure of the TPT window: $t \approx$ 30–50 Ma.
6. Onset of the Hadean dynamo: $t \approx$ 330–350 Ma.

10.3 Age of the Moon: LMO Solidification versus Formation

The measured age (4.425 ± 0.025 Ga) is that of lunar magma ocean (LMO) solidification, not formation. If the TPT is triggered at ≈ 4.467 Ga, LMO cooling (~ 40 Ma) yields a predicted solidification age:

$$t_{\mathrm{LMO}} \approx 4.467 - 0.040 = 4.427\,\mathrm{Ga}, \tag{77}$$

consistent at 0.1σ with the measured value. Prediction P8 is a formation age $>$ 4.45 Ga, directly testable by Artemis III.

11 Internal Stratification and the Seismology Program

11.1 Architecture of Concentric Layers

Each ejected sheet originates from magma at a progressively more advanced stage of Fe-Ni segregation:

$$\left(\frac{\mathrm{Fe}}{\mathrm{Si}}\right)_{\mathrm{layer}\,k} > \left(\frac{\mathrm{Fe}}{\mathrm{Si}}\right)_{\mathrm{layer}\,k+1}, \tag{78}$$

because the source magma becomes progressively depleted in siderophile elements between episodes. Density contrasts between layers $\Delta\rho_{k,k+1} \approx$ 100–200 kg m^{-3} (Garcia et al., 2019) produce potentially detectable acoustic impedance contrasts.

11.2 Preservation Window: 200–530 km

Two destructive processes bound the preservation window:

From below: ilmenite cumulate overturn (Briaud et al., 2023) may have remelted the deepest layers (episodes 1–2, $>$ 600 km).

From above: mare basalt volcanism, active until ≈ 2 Ga (Li et al., 2021; Che et al., 2021), remelted the shallowest layers ($<$ 100 km).

The surviving preservation window is therefore approximately 200–530 km deep.

11.3 Interface Depth: N=2 vs. N=3

For N=2 with equal masses, the interface between the two layers lies at:

$$d = R_{\text{Moon}} - R_1 \approx 1{,}737 - 1{,}560 \approx 177\,\text{km} \quad (N = 2). \tag{79}$$

For N=3, the primary interface lies within the 200–315 km window. Chang'e 7 seismic data will therefore allow discrimination between $N = 2$ and $N = 3$.

12 Observational Constraints and Existing Evidence

12.1 Isotopic Identity Δ^{17}O < 5 ppm

The Earth–Moon isotopic composition is nearly identical for O, Cr, and Ti, established to better than 5 ppm (Wiechert et al., 2001; Dauphas et al., 2014; Dauphas, 2017; Sossi et al., 2025). The TPT satisfies this constraint *mechanically*: at each ejection episode, material is drawn from the contemporary Hadean terrestrial mantle, with no exogenous isotopic source. No *ad hoc* isotopic mixing is required.

12.2 Iron Depletion

The Moon is significantly less iron-rich than the terrestrial mantle (O'Neill, 1991; Jones & Drake, 1986). In the TPT, this depletion is a direct consequence of the single engine: at each successive episode, the ejected mantle is more depleted in siderophiles than the previous one, as Fe-Ni segregation progressively removes iron from the source zone between episodes.

12.3 Angular Momentum

The Earth–Moon system's angular momentum budget requires a rapidly rotating proto-Earth. The value $T_{\text{rot}} = 3.5$ h is consistent with the median value from N-body accretion simulations (Agnor et al., 1999; Kokubo & Genda, 2010): rapid rotation is the norm at the end of accretion, not an exception.

12.4 Chang'e-6 Results: Preliminary Observational Support for P3 and P6

The Chang'e-6 mission returned 1,935 g of samples from the Apollo basin within the South Pole–Aitken (SPA) region in June 2024. Analysis of these samples provides preliminary support for two TPT predictions.

Norites at 4247 ± 5 Ma — the age of Episode 1. Yue et al. (2026) dated Chang'e-6 norite samples at 4247 ± 5 Ma using the Pb-Pb method. These norites are interpreted as differentiated products of the SPA impact melt pool, representing deep lunar mantle material excavated by the SPA impact. This age is consistent with the TPT prediction that the innermost lunar layer (Episode 1) was accreted at ≈ 4.5 Ga and partially reset by the SPA impact at ≈ 4.25 Ga.

Anomalously high Fe/Mn in deep olivines — a signature of Episode 1's Fe-rich material. Xu et al. (2025) report olivine fragments in Chang'e-6 samples containing significantly higher Fe and Mn concentrations than those found in the standard lunar mantle. Zinc concentrations are $\approx 100\times$ higher than expected in the lunar mantle. Although part of this enrichment is attributed to meteoritic contamination, the olivine fragments themselves formed through mixing of lunar rocks and molten asteroid material. The high Fe content in material excavated from SPA depths is *consistent* with prediction P3 (Fe/Si gradient increasing with depth) and provides preliminary support for prediction P6.

13 Falsifiable Predictions

P1 — Seismic Interface at $d \approx 200$–315 km (Priority 1)

The concentric stratification produces acoustic impedance contrasts $|R| \in [0.01, 0.04]$ between layers. For N=2, the interface is at $d \approx 177$ km. For N=3, the primary interface lies at $d \approx 200$–315 km.

Detectability with Chang'e-7 depends on the seismometer's sensitivity (target 10^{-8} m s^{-2} Hz$^{-1/2}$ at 0.1–1 Hz). This sensitivity should resolve $|R| > 0.01$, provided at least 10 events with $M > 2.5$ are recorded within the first 6 months of operation.

Falsification condition: no phase conversion detected within the 200–530 km window after deployment of at least 3 seismic stations and detection of at least 10 lunar events with $M > 2.5$.

Missions: Chang'e 7 (August 2026, South Pole seismometer), FSS (Farside Seismic Suite), LEMS (Lunar Environment Monitoring Station), Artemis III (2028–2029).

P2 — Seismic Asymmetry, Nearside vs. Farside

The second, asymmetric episode produces a sharper and deeper interface beneath the farside.

Falsification: spherical symmetry confirmed by a complete tomographic inversion.

P3 — Fe/Si Gradient Increasing with Depth

Fe/Si increases with depth in the lunar mantle, reflecting the progressive Fe-Ni depletion of the source magma between episodes. The SPA basin preferentially exposes Episode 1 material (the most Fe-rich).

Preliminary support: Chang'e-6 olivines with high Fe/Mn (Xu et al., 2025). *Missions:*

Chang'e-6 (further analyses), Artemis III.

P4 — Dynamo Delay: 290–360 Ma

The progressive dynamo emergence predicts continuous paleointensity growth from 4.55 to 4.20 Ga, with no abrupt *onset* (Tarduno et al., 2025).

Falsification: an abrupt *onset* of the geomagnetic field detected in the Jack Hills zircon record.

P5 — Instability Window: $\varepsilon \in [57°, 70°]$

The parametric elliptical instability requires ε within this range.

Test: N-body simulations of telluric proto-planet obliquity.

P6 — Polar Ejecta Signature at High Latitudes

Wobble ejects material from all latitudes of the body frame. High-latitude ejecta preferentially concentrate at the South Pole during the second, asymmetric episode. Prediction: elevated Fe/Si ratio and a distinct seismic signal at the South Pole relative to equatorial Apollo sites.

Mission: Chang'e 7 (South Pole, August 2026).

P7 — Signature in the Hf-W Chronometer

A formation duration of 3–4 weeks is the only intermediate temporal window between the giant impact (a few hours) and the synestia ($\sim$ 100 years) on the Hf-W chronometer (half-life: 8.9 Myr). The ^{182}Hf/^{180}Hf ratio in Episode 1 samples should correspond to a differentiation age of 4.467 Ga (100 $\pm$ 10 Ma after CAIs), versus 60 $\pm$ 10 Ma for the giant impact (Kleine et al., 2002).

Mission: Artemis III (2028–2029).

P8 — Formation Age > 4.45 Ga

The measured age of 4.425 Ga is that of LMO solidification, not formation (Section 10). The formation age is $\approx$ 4.467 Ga.

Test: direct dating of Episode 1 rocks exposed on the central peaks of the SPA basin (Artemis III).

P9 — Mare Basalt Compatibility with Layer 1

The melting depth of mare basalts (200–500 km, $\pm$100 km) is compatible with a source in layer 1 ($> d_{\mathrm{P1}}$), without this being assertable in the absence of more precise geochemical data.

Test: high-resolution isotopic geochemistry of Apollo and Chang'e-5 mare basalts (Li et al., 2021).

P10 — Seismic Interface at $d \approx 177$ km for $N = 2$

For $N = 2$ episodes of equal masses, the single interface between the two concentric layers lies at $d \approx 177$ km. This depth lies outside the main preservation window (200–530 km) because it has been partially remelted by mare volcanism and/or ilmenite cumulate overturn. However, partial remnants could be detected as weak reflectors or seismic velocity variations within the 150–200 km zone.

P20 — Asymptotic Regime Indicator

The actual TPT ejecta spectrum (Section 9) lies between the monokinetic and broad-spectrum bounds, with $\tilde{\sigma}_v \approx 0.05$–$0.10$. The sharp-cutoff limit provides an upper bound on L_{loss}, while the monokinetic limit provides a lower bound. The true value, accessible via Monte Carlo simulation, should lie between these bounds.

Test: Monte Carlo sampling of $\rho(v, \theta)$ (Section 9.4) compared against the analytical monokinetic and broad-spectrum limits (Section 9.7).

P21 — Numerical Closure of the Angular Momentum Budget

A Monte Carlo simulation of the ejecta spectrum (Section 9), with parameters sampled from their uncertainty ranges, should yield a closed angular momentum budget (Eq. 49) to within $\sim 5\%$.

Falsification: failure to close the angular momentum budget to within $\sim 5\%$ would indicate either an error in the distribution function $\rho(v, \theta)$ or the need for an additional angular momentum sink (e.g., tidal dissipation) not currently accounted for in the model.

P22 — Double-Well Emergence Threshold

We propose that the double-well potential governing the ejection episodes (Section 7.2.2, Appendix C) emerges only once the CMT's effective gravity has decreased by $\sim 13\%$ relative to its initial value, as a consequence of progressive Fe-Ni segregation. The double well is not present at the initial proto-Earth state.

Test: 3D SPH simulations coupling CMT dynamics with Fe-Ni segregation (Limitation L1, L2), tracking the discriminant Δ (Appendix C, Eq. 96) as a function of segregation progress.

Falsification: the absence of this threshold in such simulations, or a markedly different threshold value, would weaken the present interpretation of the ejection episodes as Kramers-type barrier crossings, though it would not by itself rule out the broader

TPT framework if an alternative ejection mechanism could be identified.

14 Validation Windows: Three Imminent Tests

14.1 Chang'e 7: Imminent Seismic Validation (August 2026)

The Chang'e 7 mission (CNSA, launch scheduled August 2026) will deploy the first seismometer at the lunar South Pole since Apollo 17 (1972). Its data will provide a direct test of P1 and P6. The prediction is pre-registered and quantified: $d \approx 200$–315 km, $|R| \in [0.01, 0.04]$. The falsification criterion is explicit.

14.2 The South Pole–Aitken Basin: Stratigraphic Revealer

The SPA basin did not create the concentric lunar stratigraphy: it *excavated* it. The layered structure predated the SPA impact at ≈ 4.25 Ga (Yue et al., 2026), encoded during TPT differentiation at ≈ 4.5 Ga. The impact is the accidental revealer of a much older internal architecture.

High-resolution 3D simulations (Wakita et al., 2026) show that the north-to-south oblique trajectory of the SPA impactor preferentially projected deep mantle ejecta toward the South Pole in a butterfly-shaped plume — precisely the Artemis III landing zone. These same simulations predict ≈ 350 m of mantle material on average at this site, excavated from depths greater than 90 km, consistent with the TPT preservation window.

14.3 Artemis III: Direct Sampling of Episode 1 Material

The Artemis III mission (NASA, planned 2028–2029) will land astronauts near the Shackleton crater at the South Pole. According to Wakita et al. (2026), surface samples at this site may include deep mantle ejecta from the SPA impact. If so, these samples will allow a direct test of P3, P7, and P8: their Fe/Si ratio, their crystallization age (≈ 4.467 Ga for primary Episode 1 material), and their isotopic signature (Δ^{17}O < 5 ppm relative to the contemporary terrestrial mantle) are all quantitative TPT predictions.

Established Result

Convergence of missions. Chang'e 7 (2026): seismology, P1, P2, P6. Artemis III (2028): geochemistry, P3, P7, P8. Two independent space agencies, one unique geographic site, one internal architecture to confirm or refute.

15 Predicted Falsification Pathways

What Would Seriously Challenge the TPT

1. **No seismic interface within 200–530 km** after adequate station coverage (P1 falsified).
2. **Spherically symmetric lunar mantle** according to complete tomography (P2 falsified).
3. **Abrupt dynamo *onset*** at $\approx$ 4.2 Ga rather than continuous growth (P4 falsified).
4. **Fe/Si of SPA mantle ejecta not higher** than terrestrial peridotite or mare basalts after correcting for impact mixing (P3 and P6 falsified).
5. **Hf-W age** incompatible with 100 ± 10 Ma after CAIs (P7 falsified).

Conversely, the TPT **does not require**:

- A specific initial rotation period other than the 2.5–4.0 h range.
- A single value of f_{cap} for the *mass budget* to be satisfied; the latter remains robust for $f_{\mathrm{cap}} \in [0.5, 0.85]$. This robustness does not, however, apply to P17 (Section 8.3), which specifically predicts $f_{\mathrm{cap}} \gtrsim 0.65$–$0.70$ as a distinguishing signature relative to the classical disk-accretion interval.
- An entirely intact ejected flow; partial fragmentation (L3) does not break the mechanism.
- A 3D SPH simulation prior to observational tests.

16 Discussion

16.1 Comparison with the Giant-Impact Model

Constraint	Giant Impact	TPT (This Work)
Isotopic identity	Requires a very specific impactor composition and mixing conditions	Satisfied mechanically by construction
Iron depletion	Requires a pre-differentiated impactor (Theia)	Direct consequence of the single engine
Crustal dichotomy	Not naturally predicted	Qualitative prediction: post-Episode 2 precession (L7)
Dynamo delay	Not predicted	Proposed: $\approx$ 350 Myr, with no free parameter
Seismic predictions	None	P1 testable by Chang'e 7 (August 2026)
Chang'e-6 Fe anomaly	Not predicted	Consistent with P3 and P6 (Xu et al., 2025; Yue et al., 2026)
Formation timescale	A few hours	3–4 weeks
Hf-W signature	60 ± 10 Ma after CAIs	100 ± 10 Ma after CAIs (P7)

A recent review concludes that there is currently no unambiguous geochemical or isotopic evidence for the role of an external impactor in the formation of the Moon (Sossi et al., 2025). This conclusion motivates the exploration of alternatives grounded in the internal dynamics of the proto-Earth.

16.2 Justification of $\alpha > 1$

Two independent mechanisms lead to a super-rigid rotation profile: differential equator–pole cooling generates a density gradient and thermal spin-up; Fe-Ni segregation lightens the CMT, and by angular momentum conservation, its inner part accelerates. The Bingham-Herschel threshold also blocks small scales, favoring a block-like profile. Numerical validation of this profile is a priority of the 3D SPH simulation.

16.3 Partial Fragmentation and Coalescence

Even if the ejected sheet fragments partially, the fragments remain within the same orbital zone and rapidly coalesce under inelastic collisions and the gravitational attraction of the proto-POB (Canup & Asphaug, 2001). The effective capture yield therefore remains high ($f_{\mathrm{cap}} \geq 0.7$). A detailed analysis (Appendix F) shows that f_{cap} has a physical lower bound of about 0.45–0.55, independent of the BH cohesion hypothesis.

16.4 Responses to the Main Objections

Detailed responses to the main objections are given in Appendix H. Here are the essential points:

"Equation (3) assumes instantaneous formation of the Earth." This is a thermodynamic bound, not a timescale. The total gravitational energy released exceeds the melting energy by a factor of 155, a result established by Solomatov (2000), Elkins-Tanton (2012), and Rubie et al. (2015).

"ε is astronomical obliquity; there is no radial Coriolis force." ε is the angle in the body's frame, defined in Section 1.5. The radial/horizontal decomposition of the Coriolis force follows directly from this geometry.

"The CMT has not been demonstrated." The CMT is the spatial expression of the $\ell = 1, m = 1$ mode of the geostrophic inverse cascade. Its quantitative validation requires a 3D SPH simulation (L1), but the theory does not depend on this for its observational tests.

"Taylor-Proudman prevents the CMT." The columns are curved by five mechanisms; their curvature generates the toroidal flow. The Taylor-Proudman constraint does not apply to the purely toroidal field (Frigaard & Nouar, 2017).

"The dynamo delay is arbitrary." It is the sum of three durations constrained by three independent observations (Section 7.3).

17 Acknowledged Limitations

L1 — Spontaneous Organization of the CMT

Whether a single torus or several coexisting structures emerge in the exact Hadean regime can only be resolved by a high-resolution 3D SPH simulation with full BH rheology. **Priority 2.**

L2 — Numerical Value of f_{cap}

The capture efficiency $f_{\mathrm{cap}} = 0.70$ is physically motivated but not derived from first principles. Its growing dependence on M_{Moon} and its gap relative to the classical disk-accretion interval (10–55%, Kokubo et al. 2000) are formalized as the testable prediction P17 (Section 8.3). The 3D SPH simulation remains the ultimate arbiter of the exact value of f_{cap} and of the validity of the orbital-concentration argument underlying P17.

L3 — Small-Scale Cohesion

The $\mathrm{Bi}_{\mathrm{KH}} \gg 1$ argument is valid at large and intermediate scales. At very small scales ($\lambda_{\mathrm{KH}} \ll R_{\mathrm{torus}}$), partial fragmentation cannot be analytically excluded. However, fragments from the same toroidal flow, ejected with the same velocity and angle, end up on nearly identical orbits and accrete onto the POB: the mass budget is not affected by fragmentation, only the fragment size.

L4 — Dynamo Delay: ± 50 Myr

The exponential core-growth model is simplified; the associated uncertainty is ± 50 Myr.

L5 — Super-Rigid Profile $\alpha > 1$

The super-rigid rotation profile ($U \propto r^{\alpha}$, $\alpha > 1$) is physically motivated by differential rotation in a partially crystallized magma undergoing rapid cooling, but has not been demonstrated under these exact Hadean conditions. This is a Level 3 hypothesis, to be validated by simulation.

L6 — Flow $\rightarrow$ POB Transition: Qualitative Status

The three-phase description of the ejected flow (ballistic expansion, orbital insertion, sticky accretion beyond R_{Roche}) is physically motivated and consistent with the calculated orders of magnitude, but remains qualitative. A 3D SPH simulation with coupled radiative cooling is required. **Priority 2b.**

L7 — Crustal Dichotomy: Qualitative Prediction

The $\approx 2:1$ thickness ratio is a qualitative prediction of the TPT. The coupled differential thermal calculation has not yet been derived analytically.

Additional context. The lunar farside has undergone at least two independent remelting episodes that must be accounted for in any quantitative model: (1) the SPA basin-forming impact at ≈ 4.25 Ga, which excavated and partially remelted the deepest Episode 1 layer on the farside (Wakita et al., 2026); and (2) prolonged mare volcanism extending until ≈ 3.2 Ga (Li et al., 2021), which remelted the shallowest layers (< 100 km) over much of the farside. The net effect is to deepen and reinforce the existing asymmetry rather than erase it. The extruded rocks exposed in the central peaks of SPA basin craters should retain a geochemical record of pre-impact Episode 1 material: their composition (Fe/Si, Δ^{17}O) is expected to match the contemporary Hadean terrestrial mantle (prediction P3).

L8 — Rossby Wave Speed in a BH Fluid

The rigorous derivation of $c^{(\mathrm{BH})}_{\mathrm{Rossby}}$ for a BH fluid on an oblate spheroid ($\varepsilon \neq 0$) remains an open problem. The channel-analogy correction (factor ≈ 40) used here strengthens rather than invalidates the synchronization argument, but awaits a rigorous derivation.

L9 — Impact Mixing in SPA Ejecta

The geochemical signature of deep mantle material excavated by the SPA impact is potentially diluted or modified by mixing with impactor material and local crust. Quantitative deconvolution of the primary Episode 1 signature from the impact-melt breccia requires detailed petrological and isotopic studies. This limitation affects predictions P3 and P6.

L10 — Deep Mantle Samples Predicted but Not Yet Collected

The TPT predicts that Episode 1 deep mantle material is exposed on the central peaks of craters within the South Pole–Aitken basin, excavated and extruded during the SPA impact at ≈ 4.25 Ga (Wakita et al., 2026). These samples are the primary targets for testing predictions P3 (Fe/Si gradient), P7 (Hf-W age), and P8 (formation age > 4.45 Ga). This prediction will be directly testable by **Chang'e 7** (seismology and remote sensing at the South Pole, August 2026) and **Artemis III** (direct sampling on the central peaks of the SPA basin, 2028–2029). However, no samples from these central peaks have yet been collected or analyzed. The theory thus makes a clear, falsifiable prediction awaiting these imminent missions. The absence of deep mantle material at the predicted locations, or an Fe/Si ratio incompatible with P3, would seriously challenge the TPT.

18 Conclusion

This work proposes that the formation of the Moon could be the product of a Triple Phase Transition within a fully molten proto-Earth rotating at ≈ 3.5 h, whose axis oscillates within $[40°, 70°]$ in its own co-rotating frame — a range supported by five convergent arguments, the first and most fundamental of which is the logically necessary absence of any tidal stabilizer. A single engine — progressive Fe-Ni segregation — simultaneously governs three transitions: rheological, mechanical, and magnetic. The result is a Moon built layer by layer in two to three massive episodes, each layer archiving in its internal structure the state of Hadean differentiation at the time of its accretion.

The parametric elliptical instability — established in fluid mechanics (Malkus, 1968; Kerswell, 2002; Lacaze et al., 2004) but never applied to lunar formation — amplifies the CMT to the bifurcation velocity within ≈ 32 hours. The generalized instability equation $\lambda_{\mathrm{global}}(\varphi, \varepsilon, t)$ is valid at all latitudes of the body frame. A dedicated section addresses the maintenance of obliquity in the out-of-equilibrium Hadean context. A chronology of three to four weeks yields three new testable predictions (P6, P7, P8).

The mass budget is consistent with no forcing: the target mass for the TPT mechanism is lower than the currently observed lunar mass, because the SPA impactor and late accretion contributed afterward. At $T_{\mathrm{rot}} = 3.5$ h, the ejectable mass per episode can reach $\approx 70\%$ of the current lunar mass, which explains why $N = 2$–3 suffices.

Preliminary observational support comes from Chang'e-6: norites at 4247 ± 5 Ma (Yue et al., 2026) and olivines with high Fe/Mn (Xu et al., 2025) are consistent with predictions P3 and P6, although not yet confirmatory.

The central prediction remains: one or more seismic interfaces within the 200–530 km window (or at ≈ 177 km for N=2), testable by Chang'e 7 in August 2026. If an interface with $|R| > 0.02$ is detected within this window, the concentric stratification receives its first strong observational support ahead of any SPH simulation. If no interface is detected under the stated falsification conditions, prediction P1 is seriously challenged and the theory will require revision — which the author explicitly accepts.

This work invites the planetary science community to evaluate the TPT on its own terms: the equations are numbered, the derivations are explicit, the hypotheses are hierarchized, the limitations are acknowledged, and the predictions are falsifiable. The TPT will be judged primarily on its observational predictions — seismic data from Chang'e 7 (August 2026) and samples from Artemis III (2028–2029) — before and independently of any complete numerical validation.

A Adiabatic Temperature Profile

With $T_{\text{surf}} \approx 3,500$ K and an adiabatic gradient $\mathrm{d}T/\mathrm{d}r \approx -0.3\,\mathrm{K\,km^{-1}}$, the central temperature reaches $\approx 5,000$ K. The solidus of peridotite at 135 GPa is $\approx 4,300$ K (Stixrude & Lithgow-Bertelloni, 2014): melting is thermodynamically stable throughout the mantle, confirming H0.

B Demonstration of the Attractor $b' = 1$

The parameter b' is defined as the ratio between the critical ejection velocity U_{crit} and the geostrophic equilibrium velocity U_{eq}:

$$b' = \frac{U_{\text{crit}}}{U_{\text{eq}}} = \frac{2\Omega \sin\varepsilon \sqrt{2|g_{\text{eff}}|R_{\text{eq,proto}}}}{|g_{\text{eff}}|}. \tag{80}$$

Evolution due to Fe-Ni segregation. The CMT density decreases as $\rho_{\text{CMT}}(t) = \rho_0(1 - \xi_{\text{Fe}}(t))$, where $\xi_{\text{Fe}}(t)$ is the fraction of segregated Fe-Ni. Effective gravity $|g_{\text{eff}}|$ is proportional to ρ_{CMT}: $|g_{\text{eff}}| \propto \rho_{\text{CMT}}$. Thus,

$$\left.\frac{\mathrm{d}b'}{\mathrm{d}t}\right|_{\text{segregation}} = \frac{1}{2}\frac{\dot{\rho_{\text{CMT}}}}{\rho_{\text{CMT}}}\, b' > 0, \tag{81}$$

since $\dot{\rho_{\text{CMT}}} < 0$. **Fe-Ni segregation increases b'.**

Evolution due to ejection. An ejection carries away a fraction $\epsilon = M_{\text{ej}}^{\max}/M_{\text{CMT}} \approx 0.1$ of the CMT mass. Angular momentum decreases as $\Delta\Omega/\Omega \approx -\epsilon$. Effective gravity decreases as $\Delta|g_{\text{eff}}|/|g_{\text{eff}}| \approx -\epsilon$ (proportional to the lost mass). Thus,

$$\left.\frac{\Delta b'}{b'}\right|_{\text{ejection}} = \frac{\Delta\Omega}{\Omega} - \frac{1}{2}\frac{\Delta|g_{\text{eff}}|}{|g_{\text{eff}}|} \approx -\epsilon + \frac{1}{2}\epsilon = -\frac{1}{2}\epsilon < 0. \tag{82}$$

Ejection decreases b'.

Stable fixed point. The two trends oppose each other: segregation increases b', ejection decreases it. The equilibrium point $b' = 1$ is a stable attractor. For $b' < 1$, the system is unstable and an ejection occurs, which decreases b' (since $\Delta b' < 0$). For $b' > 1$, the system is stable, no ejection occurs, and segregation drives b' upward toward 1 from below. The point $b' = 1$ is therefore a stable attractor.

C Complete Derivation of the Coefficient λ and the Double-Well Potential

This appendix gives the complete analytical derivation of the stability coefficient $\lambda(U)$ and of the effective potential $V(U)$, with an explicit dimensional check at every step. An earlier version of this derivation omitted a factor $1/r$ on the gravity and Coriolis terms; the corrected form is given below and used throughout the main text.

C.1 The Stability Equation

Application of the Solberg-Høiland parcel displacement formalism (Solberg, 1936; Høiland, 1941) to an oblate sphere with obliquity ε, under angular momentum conservation ($\tau_{\text{relax}}/\tau_{\text{dyn}} \approx 3 \times 10^9 \gg 1$), governs the evolution of an infinitesimal radial displacement ξ_r of a CMT parcel:

$$\ddot{\xi}_r = \lambda(U)\,\xi_r, \quad \lambda > 0 \text{ (instability)}, \quad \lambda < 0 \text{ (stable confinement)}. \tag{83}$$

The parcel conserves its specific angular momentum $L = U \cdot r$:

$$U_{\text{parcel}}(r + \xi_r) = \frac{U(r) \cdot r}{r + \xi_r} \approx U(r)\left(1 - \frac{\xi_r}{r}\right), \tag{84}$$

while the background flow at the new position, for a profile $U(r) \propto r^\alpha$, is:

$$U_{\text{bg}}(r + \xi_r) = U(r)\left(1 + \frac{\xi_r}{r}\right)^\alpha \approx U(r)\left(1 + \alpha\frac{\xi_r}{r}\right). \tag{85}$$

The velocity difference is therefore:

$$\delta U = U_{\text{parcel}} - U_{\text{bg}} = -U(r)(1 + \alpha)\frac{\xi_r}{r}. \tag{86}$$

The net radial force sums five physical contributions: effective gravity, radial Coriolis, the angular-momentum-gradient term, the Euler acceleration associated with the axial oscillation, and the tangential gravity gradient of the Maclaurin geometry. Collecting these terms and enforcing dimensional consistency (each term must carry units of s^{-2}, since $[\lambda] = [\ddot{\xi}_r/\xi_r]$) gives:

$$\boxed{\lambda(U) = \lambda_1 + \lambda_2(U) + \lambda_3(U) + \lambda_4 + \lambda_5,} \tag{87}$$

with:

$$\lambda_1 = \frac{c}{r}, \quad c = g_{\text{eff}}^{(k)} < 0 \quad \text{(effective gravity)}, \tag{88}$$

$$\lambda_2(U) = \frac{b}{r}U, \quad b = 2\Omega^{(k)} \sin \varepsilon > 0 \quad \text{(radial Coriolis force)}, \tag{89}$$

$$\lambda_3(U) = aU^2, \quad a = -\frac{1+\alpha}{r^2} < 0 \quad \text{(angular momentum gradient)}, \tag{90}$$

$$\lambda_4 = \frac{|\dot{\Omega}|\, r(\phi) \sin(\varepsilon + \phi)}{R_{\text{eq,proto}}} \quad \text{(Euler acceleration)}, \tag{91}$$

$$\lambda_5 = \frac{g_{\text{eff}}^{(0)} J_2 \sin(2\phi)}{R_{\text{eq,proto}}} \quad \text{(Maclaurin tangential gradient)}. \tag{92}$$

The angular-momentum-gradient term λ_3 is always stabilizing for any physical profile ($\alpha > -1$), consistent with the classical Rayleigh stability criterion; its amplitude ($\sim 10^{-7}$) is negligible compared with the dominant terms.

Dimensional Check

Each term in Eq. 87 has units of s^{-2}:

$[\lambda_1] = [g_{\text{eff}}]/[r] = (\text{m}/\text{s}^2)/\text{m} = \text{s}^{-2}$, $\quad [\lambda_2] = [\Omega][U]/[r] = \text{s}^{-1} \times (\text{m}/\text{s})/\text{m} = \text{s}^{-2}$,
$[\lambda_3] = [a][U]^2 = \text{m}^{-2} \times \text{m}^2/\text{s}^2 = \text{s}^{-2}$, $\quad [\lambda_4] = [\dot{\Omega}][r]/[R] = \text{s}^{-2} \times \text{m}/\text{m} = \text{s}^{-2}$,
$[\lambda_5] = [g_{\text{eff}}]/[R] = (\text{m}/\text{s}^2)/\text{m} = \text{s}^{-2}$.

The factor $1/r$ in λ_1 and λ_2 is the correction needed for consistency; it was absent from earlier drafts of this work.

C.2 Integration to the Effective Potential

By definition, the effective potential is related to the stability coefficient by $\lambda(U) = -dV/dU$. Substituting Eq. 87 and integrating with respect to U:

$$\boxed{V(U) = \frac{1+\alpha}{3r^2}U^3 - \frac{b}{2r}U^2 - \frac{c}{r}U - (\lambda_4 + \lambda_5)U + V_0,} \tag{93}$$

where V_0 is an arbitrary integration constant. Each term carries units of m/s^3, consistent with $[V] = [\lambda][U]$.

C.3 Extrema and the Double-Well Condition

The extrema of $V(U)$ satisfy $dV/dU = 0$, i.e. $\lambda(U) = 0$. Multiplying by $-r^2$ gives the quadratic equation $AU^2 + BU + C = 0$ with:

$$A = 1 + \alpha > 0, \qquad B = -br < 0, \qquad C = r|g_{\text{eff}}| - r^2(\lambda_4 + \lambda_5). \tag{94}$$

Two real roots exist if and only if the discriminant is positive:

$$\boxed{\Delta = b^2 r^2 - 4(1+\alpha) r |g_{\rm eff}| + 4(1+\alpha) r^2 (\lambda_4 + \lambda_5) > 0.} \tag{95}$$

Neglecting λ_4, λ_5 relative to $|g_{\rm eff}|/r$, this simplifies to:

$$\boxed{(2\Omega \sin \varepsilon)^2 r^2 > 4(1+\alpha)\, r\, |g_{\rm eff}|.} \tag{96}$$

When satisfied, the two roots $U_{1,2} = \left[br \pm \sqrt{\Delta}\right] / [2(1+\alpha)]$ correspond to a local minimum U_1 (confined well P_1) and a local maximum U_2 (barrier), with a second local minimum (ejective well P_2) at $U > U_2$, since the cubic term drives $V(U) \to +\infty$ as $U \to +\infty$. The barrier height is $\Delta E_{\rm barr} = V(U_2) - V(U_1)$.

C.4 Numerical Evaluation: the Double Well Is Not a Given

Table 4: Nominal parameters for the double-well condition.

Parameter	Symbol	Value
Angular velocity	Ω	4.99×10^{-4} s^{-1}
Obliquity (body frame)	ε	$55°$
Effective gravity	$\|g_{\rm eff}\|$	6.42 m/s^2
Characteristic radius	r	7.4×10^6 m
Rotation profile exponent	α	1.2

Evaluating Eq. 96 at these nominal values:

$$\text{LHS} = (2\Omega \sin \varepsilon)^2 r^2 = 3.66 \times 10^7 \text{ m}^2/\text{s}^2, \tag{97}$$

$$\text{RHS} = 4(1+\alpha) r |g_{\rm eff}| = 4.18 \times 10^8 \text{ m}^2/\text{s}^2. \tag{98}$$

Honest Reading of the Numerical Result

At nominal initial parameters, LHS $<$ RHS by roughly a factor of 11: the double-well condition is *not* satisfied. This does not undermine the TPT framework; rather, it indicates that the double well is not a property of the initial proto-Earth, but a state that must be reached dynamically. It emerges as Fe-Ni segregation reduces the CMT's effective gravity $|g_{\rm eff}|$, since $|g_{\rm eff}|(t) \propto \rho_{\rm CMT}(t)$ (Eq. 24) and $\rho_{\rm CMT}(t)$ decreases monotonically with segregation.

The threshold is reached when:

$$|g_{\mathrm{eff}}|(t) < \frac{\Omega^2 r \sin^2 \varepsilon}{1+\alpha} \approx 5.62 \mathrm{~m/s^2}, \tag{99}$$

i.e. a reduction of $|g_{\mathrm{eff}}|$ by only $\sim 13\%$ relative to its initial value of $6.42 \mathrm{~m/s^2}$ is needed for the system to enter the double-well regime.

P22 — Double-Well Emergence Threshold

We propose that the double-well potential emerges once the CMT's effective gravity has decreased by $\sim 13\%$ relative to its initial value ($|g_{\mathrm{eff}}| \lesssim 5.6 \mathrm{~m/s^2}$), as a consequence of progressive Fe-Ni segregation. This threshold could be tested in 3D SPH simulations coupling CMT dynamics with Fe-Ni segregation; its absence in such simulations, or a markedly different threshold value, would weaken the present interpretation of the ejection episodes as Kramers-type barrier crossings.

Once the well is established, the barrier height $\Delta E_{\mathrm{barr}}^{(k)}$ is expected to decrease with each subsequent episode as segregation continues, offering a possible account for why the number of episodes naturally settles around $N = 2$–3, rather than being imposed externally.

C.5 The Kramers Crossing Rate

Once $\Delta > 0$, the Kramers crossing rate (Kramers, 1940) applies (restating Eq. 31 from Section 7.2.2):

$$\Gamma_{\mathrm{Kramers}} = \frac{\omega_0 \omega_b}{2\pi \gamma_{\mathrm{diss}}} \exp\left(-\frac{\Delta E_{\mathrm{barr}}}{k_B T_{\mathrm{eff}}}\right), \tag{100}$$

where $\omega_0 = \sqrt{|V''(U_{P_1})|}$ and $\omega_b = \sqrt{|V''(U_{\mathrm{peak}})|}$ are the oscillation frequencies at the bottom of the well and the top of the barrier respectively, and γ_{diss} is the dissipation rate. With $N_{\mathrm{opp}} \sim 10^9$–$10^{10}$ stochastic opportunities (CMEs, impacts) available once the well is established, the probability of no crossing at all over the relevant timescale is negligible.

Summary of the Derivation

The argument proceeds as follows: (1) start from the Solberg-Høiland stability equation; (2) write $\lambda(U)$ as the sum of five dimensionally consistent physical terms; (3) integrate to obtain the cubic effective potential $V(U)$; (4) the discriminant of the resulting quadratic determines whether a double well exists; (5) at nominal initial parameters it does not, but the condition is satisfied dynamically once Fe-Ni segregation has reduced $|g_{\mathrm{eff}}|$ by $\sim 13\%$. Under this reading, the double-well potential is a consequence of the proposed TPT dynamics rather than an assumption introduced to fit the conclusion.

D Demonstration of the $\mathrm{Bi}_{\mathrm{KH}}$ Criterion

The Bingham-Kelvin-Helmholtz number compares the yield stress to the KH shear stress:

$$\mathrm{Bi}_{\mathrm{KH}} = \frac{\tau_y\, k}{\rho\, \sigma_{\mathrm{KH}}\, \Delta U}, \tag{101}$$

where k is a geometric factor (~ 1), $\sigma_{\mathrm{KH}} \sim \Delta U/\lambda$ is the KH growth rate, and λ is the perturbation wavelength. Thus,

$$\mathrm{Bi}_{\mathrm{KH}} \sim \frac{\tau_y\, \lambda}{\rho\, \Delta U^2}. \tag{102}$$

For the CMT: $\tau_y \sim 10^2$–10^4 Pa, $\lambda \sim R_{\mathrm{torus}} \sim 10^7$ m, $\rho \sim 3,000\ \mathrm{kg\,m^{-3}}$, $\Delta U \sim U_{\mathrm{crit}} \sim 10^4\ \mathrm{m\,s^{-1}}$. One obtains $\mathrm{Bi}_{\mathrm{KH}} \sim 10^{-1}$–$10^1$, i.e. $\mathcal{O}(1)$ or more. Cohesion is maintained through three reinforcements: (1) atmospheric pressure $P \sim 10^5$–10^7 Pa, (2) internal crystalline network multiplying resistance by a factor of 3–10, (3) Rayleigh-Taylor suppression ($\rho_{\mathrm{flow}}/\rho_{\mathrm{atm}} \sim 200$).

E Derivation of the Elsasser Number and B_{crit}

By balancing the Coriolis force $F_C \sim 2\rho\Omega U$ and the Lorentz force $F_L \sim B^2/(\mu L)$, with $\eta \sim UL$ (magnetic Reynolds number of order unity at the dynamo threshold):

$$B^2 \sim 2\mu\rho\Omega\eta, \qquad \Rightarrow \qquad \Lambda \equiv \frac{B^2}{\rho\mu\eta\Omega} \sim 2, \tag{103}$$

i.e. $\Lambda \sim \mathcal{O}(1)$. Using $\eta = (\mu_0\sigma)^{-1}$ with $\sigma \approx 5 \times 10^5\ \mathrm{S\,m^{-1}}$, $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{H\,m^{-1}}$, $\rho \approx 10^4\ \mathrm{kg\,m^{-3}}$, $\Omega \approx 4.99 \times 10^{-4}\ \mathrm{rad\,s^{-1}}$, one obtains $B_{\mathrm{crit}} \approx 6\,\mu\mathrm{T}$.

F Fe-Ni Stokes Sedimentation Timescale

Stokes sedimentation velocity for an Fe-Ni droplet of radius $r \approx 5$ cm in molten silicate:

$$v_{\mathrm{Stokes}} = \frac{2}{9}\frac{(\rho_{\mathrm{Fe}} - \rho_{\mathrm{sil}})\, g_{\mathrm{eff}}^{(0)}\, r^2}{\eta} \approx \frac{2}{9}\frac{3,300 \times 6.42 \times (0.05)^2}{0.1} \approx 118\,\mathrm{m\,s^{-1}}. \tag{104}$$

Over a characteristic sedimentation distance $\ell \approx 30$ km: $\tau_{\mathrm{Stokes}} = \ell/v_{\mathrm{Stokes}} \approx 254$ s. Robust for $r \in [1, 10]$ cm and $\eta \in [0.05, 1]$ Pa s.

G Accretion of Magmatic Ejecta: Why Perfect Cohesion Is Not Required

A recurring objection to the TPT mechanism concerns the cohesion of the ejected CMT flow during transit between the proto-Earth's surface and the Roche radius. This appendix demonstrates that, even without perfect cohesion of the flow, the physical properties of the ejected magma and orbital mechanics guarantee accretion into a single body.

G.1 Magma Is Not Dust

The ejected material is silicate magma at $T \approx 3,000$–$4,000$ K. Its dynamic viscosity is $\eta \approx 1$–100 Pa s (Giordano et al., 2008) — comparable to hot honey. Unlike solid rock fragments or a gas cloud, high-temperature magma does not disperse: it deforms and stretches, but remains dense and cohesive at the macroscopic scale. Two sticky magma blobs at low relative velocity do not bounce off each other — they fuse. Viscous coalescence replaces mechanical cohesion as the dominant aggregation mechanism once the flow fragments.

G.2 Orbital Concentration

Ejecta from a given TPT episode share, to first order, the same initial orbital energy budget. By conservation of energy and angular momentum, they populate a band of neighboring orbits rather than dispersing isotropically over 4π steradians. This orbital concentration is fundamentally different from an isotropic explosion: it is the same physical principle that produces accretion disks from tidally disrupted bodies.

The CMT's toroidal geometry reinforces this effect: ejection occurs preferentially within the oblique intertropical band $|\phi| < 30^\circ$, concentrating ejecta within a limited azimuthal solid angle. The resulting debris field is geometrically thin, maximizing mutual encounter rates.

G.3 Beyond the Roche Radius: Accretion Is Inevitable

Beyond the Roche radius $R_{\text{Roche}} = 2.456 R_\oplus$, the self-gravity of the ejecta cloud overcomes tidal disruption. For co-orbital magma fragments of total mass $M_{\text{cloud}} \approx 2$–$4 \times 10^{22}$ kg, the gravitational accretion timescale is:

$$\tau_{\text{acc}} \approx \frac{1}{\sqrt{G\rho_{\text{cloud}}}} \approx 10^3\text{–}10^5 \text{ years}, \tag{105}$$

far shorter than the thermal lifetime of the magma ($\tau_{\text{cool}} \sim 10^6$–$10^7$ years for a Moon-sized body, Elkins-Tanton 2008).

During this accretion window, the magma remains above its solidus: every collision is plastic, not elastic. There is no fragmentation cascade, no loss by bouncing. The capture efficiency f_{cap} therefore has a physical lower bound independent of the BH cohesion hypothesis:

$$f_{\text{cap}}^{\text{min}} \approx \frac{M_{\text{cloud}}(r > R_{\text{Roche}})}{M_{\text{ej}}^{\text{total}}} \gtrsim 0.45\text{–}0.55, \tag{106}$$

the lost fraction ($\approx$ 45–55%) corresponding to ejecta that fall back onto the proto-Earth or escape on hyperbolic orbits.

G.4 Implication for the Theory

The capture efficiency $f_{\text{cap}} = 0.70$ adopted in the main text is therefore not a free parameter that the theory must defend at all costs. Even in the pessimistic scenario where the CMT flow fragments completely upon ejection, orbital mechanics and magma viscosity guarantee accretion of the co-orbital debris cloud beyond the Roche radius.

The critical question shifts from "does the flow remain cohesive?" to "what fraction of the ejecta reaches the Roche radius?" — a question that depends on the velocity distribution of the ejecta, not on BH rheology. This reformulation makes the theory more robust: cohesion is advantageous, but not necessary.

H Detailed Responses to Objections

Objection 1 — Equation (3) assumes instantaneous formation. Equation (3) is an *energetic thermodynamic bound*, not a formation timescale. It asks the question: does the total gravitational energy released during accretion, however slow, exceed the energy required to melt the mantle? The answer, established by Solomatov (2000), Elkins-Tanton (2012), and Rubie et al. (2015), is yes, by a factor of 155. The $\approx$ 30 Myr core-formation timescale (Kleine et al., 2002) is explicitly incorporated into the TPT chronology as Δt_{ej} (Section 7.3).

Objection 2 — ε is astronomical obliquity. ε is the angle in the body's frame between $\mathbf{\Omega}$ and the principal axis of inertia of the Maclaurin spheroid — an internal geometric quantity with no reference to the ecliptic plane. The decomposition of the Coriolis force into radial and horizontal components (Eqs. 16–17) follows directly from this geometry.

Objection 3 — The CMT has not been demonstrated. The CMT is the spatial expression of the $\ell = 1$, $m = 1$ mode of the geostrophic inverse cascade at the Rhines scale in a rapidly rotating oblate body with $\varepsilon > 45°$. Four independent physical effects converge on the band $|\phi| < 30°$ as an attractor (Section 3). Quantitative validation requires a 3D SPH simulation (Priority 2), but the theory does not depend on this for its observational tests.

Objection 4 — Taylor-Proudman prevents the CMT. The columns exist but are curved by five mechanisms (Section 5). Their curvature generates the toroidal flow rather than inhibiting it. The Taylor-Proudman constraint does not apply to the purely toroidal field (Frigaard & Nouar, 2017).

Objection 5 — The dynamo delay is arbitrary. It is the sum of three durations constrained by three independent observations: (1) the Hf-W chronometer (Kleine et al., 2002; Yin et al., 2002), (2) the Jack Hills zircons (Wilde et al., 2001; Valley et al., 2002), (3) Tarduno et al. (2025).

I List of Abbreviations

BH Bingham-Herschel, rheological law.

BiKH Bingham-Kelvin-Helmholtz number.

CME Coronal Mass Ejection.

CMT Coherent Magmatic Torus.

FSS Farside Seismic Suite.

Hf-W Hafnium-tungsten chronometer.

LHB Late Heavy Bombardment ($\approx$ 3.9 Ga).

LEMS Lunar Environment Monitoring Station.

LMO Lunar Magma Ocean.

POB Proto-Orbital Body.

SPA South Pole–Aitken (basin).

SPH Smoothed Particle Hydrodynamics.

TPT Triple Phase Transition.

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